

D4-branes wrapped on topological disks from matter-coupled $F(4)$ gauged supergravity

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Abstract

We study a number of supersymmetric $AdS_4 \times \Sigma$ solutions with Σ being a topological disk with non-trivial $U(1)$ holonomy on the boundary or a “half-spindle” from matter-coupled $F(4)$ gauged supergravity. The gauged supergravity is coupled to three vector multiplets with $SO(3) \times SO(3)$ gauge group. The resulting solutions preserve eight supercharges and $SO(2) \times SO(2)$ or $SO(2)_{\text{diag}}$ symmetries and are expected to be dual to $N = 2$ SCFTs in three dimensions arising from compactifications of five-dimensional $N = 2$ SCFT on a half-spindle. All of the solutions lie within the $U(1) \times U(1)$ subsector of the matter-coupled $F(4)$ gauged supergravity that can be embedded in massive type IIA theory. After uplifted to ten dimensions, the solutions can be interpreted as a system of D4-D8-branes wrapped on Σ . Some of the solutions are asymptotic to a locally AdS_6 geometry and can be identified as codimension-2 conformal defects within the $N = 2$ SCFT in five dimensions. For these solutions, the circle inside Σ decompactifies in the AdS_6 limit rendering the holographic free energy infinite. There also exist solutions with finite free energy in the dual three-dimensional SCFTs. In addition, we show that the results both extend the previously known solutions and provide a novel class of solutions.

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1 Introduction

One of the most interesting aspects of the AdS/CFT correspondence [1, 2, 3] is to provide a holographic description of strongly coupled superconformal field theories (SCFTs) compactified on a compact manifold. In many cases, this leads to a number of new SCFTs in lower dimensions. Holographic solutions realizing the compactification on a Riemann surface of $N = 4$ SYM and $N = (2, 0)$ SCFT

in six dimensions have been studied long ago in [4]. The results have been extended to more general solutions interpolating between AdS_{m+n} and $AdS_m \times M^n$ solutions within gauged supergravity in $m+n$ dimensions by a large number of previous works, see [6]-[30] for an incomplete list. All of these solutions rely on an important mechanism of topological twist which is vital for preserving some part of the original supersymmetry.

It turns out that there are other mechanisms in which supersymmetry can be unbroken for solutions of the form $AdS_n \times \Sigma$ with Σ being a two-dimensional compact manifold with one and two orbifold singularities. These new mechanisms have been found in [31] and [32], see also [33]-[44] for extensions to higher-dimensional orbifolds. For the case of two singularities, Σ is topologically a two-sphere with orbifold singularities at the poles [45, 46, 47, 48, 49, 50, 51] while, in the case of one singularity, Σ is a topological disk with non-trivial $U(1)$ holonomy on the boundary and an orbifold singularity at the origin [52, 53, 54, 55, 56, 57]. The latter is sometimes called a “half-spindle”, see [32] for an explicit illustration. All of these solutions lead to a holographic description of higher-dimensional SCFTs compactified on a spindle or a half-spindle resulting in a large number of new SCFTs in lower dimensions. In many cases, the solutions from gauged supergravity can be uplifted to ten or eleven dimensions leading to various brane configurations realizing these new SCFTs in the context of string/M-theory.

In this work, we are interested in supersymmetric $AdS_4 \times \Sigma$ solutions from matter-coupled $N = (1, 1)$ or $F(4)$ gauged supergravity in six dimensions constructed in [58] and [59]. Since the solutions with Σ being a spindle have already been found in [50], we will focus on the case of Σ being a topological disk. A similar solution in pure $F(4)$ gauged supergravity [60] has also appeared in [54], see also [61] for solutions describing D4-branes wrapped on various supersymmetric cycles. In this paper, we will extend this result to the matter-coupled $F(4)$ gauged supergravity coupled to three vector multiplets with $SO(3) \times SO(3)$ gauge group. A number of holographic solutions from this gauged supergravity have been extensively studied in [62, 63, 64, 65, 66, 67]. In the present work, we will add to this collection supersymmetric $AdS_4 \times \Sigma$ solutions for Σ being a half-spindle. A number of possible solutions of this type have been classified in [43]. However, apart from the solution found in [54], most of these solutions have not been constructed to date. In addition to providing explicit solutions within this classification, the present results lead to new solutions that do not seem to lie in the space of solutions given in [43]. Moreover, all the solutions found here lie within the $U(1) \times U(1)$ truncation of the matter-coupled $F(4)$ gauged supergravity that can in turn be embedded in massive type IIA theory. Accordingly, the solutions can be uplifted to massive type IIA theory in which the interpretation in terms of D4-D8-brane systems can be given.

The paper is organized as follows. In section 2, we give a brief review of six-dimensional $N = (1, 1)$ gauged supergravity coupled to vector multiplets. Supersymmetric $AdS_4 \times \Sigma$ solutions with $SO(2) \times SO(2)$ and $SO(2)_{\text{diag}}$ symmetries

are considered in sections 3 and 4, respectively. Some conclusions and comments are given in section 5. We also include more details on various parts of the analysis in appendices. In appendix A, all bosonic field equations for the matter-coupled $F(4)$ gauged supergravity under consideration here are given. The derivations of solutions with $SO(2) \times SO(2)$ and $SO(2)_{\text{diag}}$ symmetries are presented in appendices B and C, respectively. In appendix D, we give some relations between the solutions given here and those involving spindles studied in [50]. We also show that some of the solutions lie in the solution space given in [43] but some do not indicating new solutions. Finally, in appendix E, we collect relevant formulae to uplift six-dimensional solutions to massive type IIA theory and needed formula for computing holographic free energy in the three-dimensional SCFTs dual to the $AdS_4 \times \Sigma$ solutions.

2 Matter-coupled $F(4)$ gauged supergravity

We first review the general structure of matter-coupled $F(4)$ gauged supergravity in six dimensions. All the conventions are mostly the same as [58] and [59], to which we refer for more details. In this paper, we will mainly present relevant formulae for obtaining supersymmetric $AdS_4 \times \Sigma$ solutions. The field content of the half-maximal supergravity coupled to n vector multiplets is collectively given by

$$(e_{\hat{\mu}}^{\hat{\mu}}, \psi_{\mu}^A, A_{\mu}^{\Lambda}, B_{\mu\nu}, \chi^A, \lambda_A^I, \sigma, \phi^{\alpha I}). \quad (2.1)$$

The metric signature is $(-++++)$ with spacetime and tangent space indices denoted respectively by $\mu, \nu = 0, \dots, 5$ and $\hat{\mu}, \hat{\nu} = 0, \dots, 5$. The bosonic fields are given by the graviton $e_{\hat{\mu}}^{\hat{\mu}}$, a two-form field $B_{\mu\nu}$, $(4+n)$ vector fields $A_{\mu}^{\Lambda} = (A_{\mu}^{\alpha}, A_{\mu}^I)$, the dilaton σ , and $4n$ scalars $\phi^{\alpha I}$. The fermionic fields are given by two gravitinos ψ_{μ}^A , two spin-1/2 fields χ^A and $2n$ gauginos λ_A^I . Indices $\alpha = (0, i) = 0, 1, 2, 3$ are split into $SU(2)_R \sim USp(2)_R \sim SO(3)_R$ R-symmetry singlet (0) and adjoint index ($i = 1, 2, 3$) while $I = 1, 2, \dots, n$ denotes the n vector multiplets. The $SU(2)_R$ fundamental indices $A, B, \dots = 1, 2$ are raised and lower by ϵ^{AB} and ϵ_{AB} with the convention $T^A = \epsilon^{AB} T_B$ and $T_A = T^B \epsilon_{BA}$.

Including the dilaton, there are $4n + 1$ scalar fields described by $\mathbb{R}^+ \times SO(4, n) / (SO(4) \times SO(n))$ coset with the $\mathbb{R}^+$ factor corresponding to the dilaton. The $4n$ vector multiplet scalars can be parametrized by a coset representative $L^{\Lambda}_{\underline{\Sigma}}$ transforming under the global $SO(4, n)$ and local $SO(4) \times SO(n)$ symmetries respectively by left and right multiplications with indices $\Lambda, \underline{\Sigma} = 0, \dots, n+3$. We can also split the index $\underline{\Sigma}$ transforming under the local $SO(4) \times SO(n)$ as $\underline{\Sigma} = (\alpha, I) = (0, i, I)$. Accordingly, the coset representative can be written as

$$L^{\Lambda}_{\underline{\Sigma}} = (L^{\Lambda}_{\alpha}, L^{\Lambda}_I). \quad (2.2)$$

The inverse of $L^\Lambda_{\underline{\Sigma}}$ will be denoted by $(L^{-1})^\Sigma_{\Lambda} = ((L^{-1})^\alpha_{\Lambda}, (L^{-1})^I_{\Lambda})$. $SO(4, n)$ indices $\Lambda, \Sigma, \dots$ will be raised and lowered by the invariant tensor

$$\eta_{\Lambda\Sigma} = \eta^{\Lambda\Sigma} = (\delta_{\alpha\beta}, -\delta_{IJ}). \quad (2.3)$$

We are interested in gauging a compact subgroup of $SO(4, n)$ of the form $G = SO(3) \times G_c \subset SO(4) \times SO(n)$. The $SO(3)$ factor is identified with the R-symmetry $SO(3)_R \sim SU(2)_R$ and gauged by three vector fields A_μ^i within the gravity multiplet. On the other hand, G_c is a compact subgroup of $SO(n)$ gauged by the vector fields in vector multiplets with $\dim(G_c) \leq n$. In addition to the gauging of $G \subset SO(4, n)$, there is a massive deformation of the two-form field relevant for the existence of supersymmetric AdS_6 vacua. With both of these deformations taken into account, the Lagrangian for $N = (1, 1)$ gauged supergravity can be written as

$$\begin{aligned} e^{-1}\mathcal{L} = & \frac{1}{4}R - \partial_\mu\sigma\partial^\mu\sigma - \frac{1}{4}P_\mu^{I\alpha}P^\mu_{I\alpha} - \frac{1}{8}e^{-2\sigma}\mathcal{N}_{\Lambda\Sigma}\hat{F}_{\mu\nu}^\Lambda\hat{F}^{\Sigma\mu\nu} - \frac{3}{64}e^{4\sigma}H_{\mu\nu\rho}H^{\mu\nu\rho} \\ & - V - \frac{1}{64}e^{-1}\epsilon^{\mu\nu\rho\sigma\lambda\tau}B_{\mu\nu}\left(\eta_{\Lambda\Sigma}\hat{F}_{\rho\sigma}^\Lambda\hat{F}_{\lambda\tau}^\Sigma + mB_{\rho\sigma}\hat{F}_{\lambda\tau}^\Lambda\delta_{\Lambda 0} + \frac{1}{3}m^2B_{\rho\sigma}B_{\lambda\tau}\right) \end{aligned} \quad (2.4)$$

with $e = \sqrt{-g}$. The various field strength tensors appearing in the above Lagrangian are defined by

$$\hat{F}_{\mu\nu}^\Lambda = F_{\mu\nu}^\Lambda - mB_{\mu\nu}\delta_0^\Lambda, \quad (2.5)$$

$$F_{\mu\nu}^\Lambda = \frac{1}{2}(\partial_\mu A_\nu^\Lambda - \partial_\nu A_\mu^\Lambda - f_\Sigma{}^\Lambda{}_\Gamma A_\mu^\Sigma A_\nu^\Gamma), \quad (2.6)$$

$$H_{\mu\nu\rho} = \frac{1}{3}(\partial_\mu B_{\nu\rho} + \partial_\nu B_{\rho\mu} + \partial_\rho B_{\mu\nu}). \quad (2.7)$$

The constant m corresponds to the aforementioned massive deformation while $f_\Sigma{}^\Lambda{}_\Gamma$ are related to the structure constants of the gauge group with the corresponding gauge algebra given by

$$[T_\Lambda, T_\Sigma] = f^\Gamma{}_{\Lambda\Sigma}T_\Gamma. \quad (2.8)$$

In this equation, we have denoted the gauge generators in the fundamental representation of $SO(4, n)$ by T_Λ . Consistency of the gaugings with supersymmetry also requires

$$f_{[\Lambda\Sigma\Gamma]} = 0. \quad (2.9)$$

For $SO(3) \times G_c$ gauge group, $f_{\Lambda\Sigma\Gamma}$ takes the form of

$$f_{\Lambda\Sigma\Gamma} = (g_1\epsilon_{ijk}, g_2C_{IJK}) \quad (2.10)$$

with g_1 and g_2 being gauge coupling constants. C_{IJK} are structure constants of G_c .

The vielbein on $SO(4, n)/(SO(4) \times SO(n))$ appearing in the scalar kinetic term can be obtained from the left-invariant one-form

$$\Omega_\mu^\Lambda{}_\Sigma = (L^{-1})^\Lambda{}_\Pi D_\mu L^\Pi{}_\Sigma \quad \text{with} \quad D_\mu L^\Lambda{}_\Sigma = \partial_\mu L^\Lambda{}_\Sigma - f_\Gamma{}^\Lambda{}_\Pi A_\mu^\Gamma L^\Pi{}_\Sigma \quad (2.11)$$

via

$$P_\mu^I{}_\alpha = (P_\mu^I{}_0, P_\mu^I{}_i) = (\Omega_\mu^I{}_0, \Omega_\mu^I{}_i). \quad (2.12)$$

The other components of $\Omega_\mu^\Lambda{}_\Sigma$ are identified as the $SO(4) \times SO(n)$ composite connections $(\Omega_\mu^{ij}, \Omega_\mu^{i0}, \Omega_\mu^{IJ})$.

The symmetric scalar matrix $\mathcal{N}_{\Lambda\Sigma}$, appearing in the vector kinetic term, is defined by

$$\mathcal{N}_{\Lambda\Sigma} = L_{\Lambda\alpha}(L^{-1})^\alpha{}_\Sigma - L_{\Lambda I}(L^{-1})^I{}_\Sigma = (\eta L L^T \eta)_{\Lambda\Sigma}. \quad (2.13)$$

The scalar potential is given by

$$\begin{aligned} V = & -e^{2\sigma} \left[\frac{1}{36} A^2 + \frac{1}{4} B^i B^i + \frac{1}{4} (C^I{}_i C_{Ii} + 4 D^I{}_i D_{Ii}) \right] + m^2 e^{-6\sigma} \mathcal{N}_{00} \\ & - m e^{-2\sigma} \left[\frac{2}{3} A L_{00} - 2 B^i L_{0i} \right] \end{aligned} \quad (2.14)$$

with various components of fermion-shift matrices defined by

$$\begin{aligned} A &= \varepsilon^{ijk} K_{ijk}, & B^i &= \varepsilon^{ijk} K_{jk0}, \\ C_I{}^i &= \varepsilon^{ijk} K_{jIk}, & D_{Ii} &= K_{0Ii}. \end{aligned} \quad (2.15)$$

The “boosted” structure constants are in turn given by

$$K_{ij\alpha} = f_\Lambda{}^\Gamma{}_\Sigma L^\Lambda{}_i (L^{-1})_{j\Gamma} L^\Sigma{}_\alpha, \quad (2.16)$$

$$K_{\alpha Ii} = f_\Lambda{}^\Gamma{}_\Sigma L^\Lambda{}_\alpha (L^{-1})_{I\Gamma} L^\Sigma{}_i. \quad (2.17)$$

Supersymmetry transformations of fermionic fields are given by

$$\begin{aligned}
\delta\psi_{\mu A} = & D_\mu \epsilon_A - \frac{1}{24} [Ae^\sigma + 6me^{-3\sigma}(L^{-1})_{00}] \epsilon_{AB} \Gamma_\mu \epsilon^B \\
& - \frac{1}{8} [B_i e^\sigma - 2me^{-3\sigma}(L^{-1})_{i0}] \Gamma_7 \Gamma_\mu \sigma_{AB}^i \epsilon^B \\
& + \frac{i}{16} e^{-\sigma} \left[(L^{-1})_{0\Lambda} \widehat{F}_{\nu\lambda}^\Lambda \Gamma_7 \epsilon_{AB} + (L^{-1})_{i\Lambda} F_{\nu\lambda}^\Lambda \sigma_{AB}^i \right] (\Gamma_\mu^{\nu\lambda} - 6\delta_\mu^\nu \Gamma^\lambda) \epsilon^B \\
& + \frac{i}{32} e^{2\sigma} H_{\nu\lambda\rho} \Gamma_7 (\Gamma_\mu^{\nu\lambda\rho} - 3\delta_\mu^\nu \Gamma^{\lambda\rho}) \epsilon_A,
\end{aligned} \tag{2.18}$$

$$\begin{aligned}
\delta\chi_A = & \frac{1}{2} \Gamma^\mu \partial_\mu \sigma \epsilon_{AB} \epsilon^B + \frac{1}{24} [Ae^{-\sigma} - 18me^{-3\sigma}(L^{-1})_{00}] \epsilon_{AB} \epsilon^B \\
& - \frac{1}{8} [B_i e^\sigma + 6me^{-3\sigma}(L^{-1})_{i0}] \Gamma_7 \sigma_{AB}^i \epsilon^B \\
& + \frac{i}{16} e^{-\sigma} \left[(L^{-1})_{0\Lambda} \widehat{F}_{\mu\nu}^\Lambda \Gamma_7 \epsilon_{AB} - (L^{-1})_{i\Lambda} F_{\mu\nu}^\Lambda \sigma_{AB}^i \right] \Gamma^{\mu\nu} \epsilon^B \\
& - \frac{i}{32} e^{2\sigma} H_{\nu\lambda\rho} \Gamma_7 \Gamma^{\nu\lambda\rho} \epsilon_A,
\end{aligned} \tag{2.19}$$

$$\begin{aligned}
\delta\lambda_A^I = & P_\mu^I \Gamma^\mu \sigma_{AB}^i \epsilon^B + P_\mu^I \Gamma_7 \Gamma^\mu \epsilon_{AB} \epsilon^B - e^\sigma (2i\Gamma_7 D^I{}_i + C^I{}_i) \sigma_{AB}^i \epsilon^B \\
& + 2me^{-3\sigma}(L^{-1})^I{}_0 \Gamma_7 \epsilon_{AB} \epsilon^B - \frac{i}{2} e^{-\sigma} (L^{-1})^I{}_\Lambda F_{\mu\nu}^\Lambda \Gamma^{\mu\nu} \epsilon_A.
\end{aligned} \tag{2.20}$$

In these equations, $\sigma^{iA}{}_B$ are the usual Pauli matrices and $\Gamma_\mu = e_\mu^{\hat{\mu}} \Gamma_{\hat{\mu}}$ in which $\Gamma_{\hat{\mu}}$ are six-dimensional spacetime gamma matrices satisfying the $SO(1, 5)$ Clifford algebra

$$\{\Gamma_{\hat{\mu}}, \Gamma_{\hat{\nu}}\} = 2\eta_{\hat{\mu}\hat{\nu}} \mathbb{1}_8 \quad \text{with} \quad \eta_{\hat{\mu}\hat{\nu}} = \text{diag}(-1, 1, 1, 1, 1, 1). \tag{2.21}$$

The chirality matrix is defined by $\Gamma_7 = i\Gamma^0\Gamma^1\Gamma^2\Gamma^3\Gamma^4\Gamma^5$ with $\Gamma_7^2 = -\mathbb{1}_8$ and $\mathbb{1}_n$ being an $n \times n$ identity matrix. The covariant derivative of ϵ_A takes the form

$$D_\mu \epsilon_A = \partial_\mu \epsilon_A + \frac{1}{4} \omega_\mu^{\nu\rho} \Gamma_{\nu\rho} \epsilon_A + Q_{\mu AB} \epsilon^B \tag{2.22}$$

in which the composite connection is defined by

$$Q_{\mu AB} = \frac{i}{2} \sigma_{AB}^i \left(\frac{1}{2} \varepsilon_{ijk} \Omega_\mu^{jk} - i\Gamma_7 \Omega_{\mu i 0} \right). \tag{2.23}$$

To give an explicit parametrization of the $SO(4, n)/(SO(4) \times SO(n))$ scalar manifold, we can choose the coset representative of the form

$$L = e^{\phi^{\alpha I} Y_{\alpha I}}. \tag{2.24}$$

The $SO(4, n)$ non-compact generators can be written in terms of $GL(n+4, \mathbb{R})$ matrices

$$(e^{\Lambda\Sigma})_{\Gamma\Pi} = \delta_\Gamma^\Lambda \delta_\Pi^\Sigma, \quad \Lambda, \Sigma, \dots = 0, 1, \dots, n+3 \tag{2.25}$$

as

$$Y_{\alpha I} = e^{\alpha, I+3} + e^{I+3, \alpha}. \quad (2.26)$$

In this paper, we are interested in $F(4)$ gauged supergravity with the $SO(3) \times SO(3) \sim SU(2) \times SU(2)$ gauge group obtained by coupling the gravity multiplet to $n = 3$ vector multiplets. As previously mentioned, the first $SO(3)$ is the R-symmetry gauged by A_μ^i , and the second one is gauged by A_μ^I , $I = 1, 2, 3$, from the three vector multiplets. Choosing $C_{IJK} = \varepsilon_{IJK}$, we have the following components of the structure constants

$$f_{\Lambda\Sigma\Gamma} = (g_1 \varepsilon_{ijk}, g_2 \varepsilon_{IJK}). \quad (2.27)$$

Explicitly, $SO(3)_R$ and $SO(3)$ are generated respectively by

$$SO(3)_R : \quad J^{ij} = e^{ji} - e^{ij}, \quad i, j = 1, 2, 3, \quad (2.28)$$

$$SO(3) : \quad \tilde{J}^{IJ} = e^{J+3, I+3} - e^{I+3, J+3}, \quad I, J = 1, 2, 3. \quad (2.29)$$

Various types of holographic solutions from this gauged supergravity have been studied in a number of previous works [62, 63, 64, 65, 66, 67]. In particular, there are two supersymmetric AdS_6 vacua preserving full supersymmetry with $SO(3) \times SO(3)$ and $SO(3)_{\text{diag}}$ symmetries. These vacua can be found by considering the $SO(3)_{\text{diag}}$ singlet scalars consisting of the dilaton σ and another scalar from $SO(4, 3)/(SO(4) \times SO(3))$. The latter is given by the following coset representative

$$L = e^{\phi(Y_{11} + Y_{22} + Y_{33})}. \quad (2.30)$$

The AdS_6 vacuum with $SO(3) \times SO(3)$ symmetry is given by

$$\phi = 0, \quad \sigma = \frac{1}{4} \ln \left[\frac{3m}{g_1} \right], \quad V_0 = -20m^2 \left(\frac{g_1}{3m} \right)^{\frac{3}{2}}. \quad (2.31)$$

We can also choose $g_1 = 3m$ or equivalently shifting the value of σ such that $\sigma = 0$ at the vacuum. V_0 denotes the cosmological constant at the vacuum. The other AdS_6 vacuum with $SO(3)_{\text{diag}}$ symmetry reads

$$\begin{aligned} \phi &= \frac{1}{2} \ln \left[\frac{g_1 + g_2}{g_2 - g_1} \right], \quad \sigma = \frac{1}{4} \ln \left[\frac{3m \sqrt{g_2^2 - g_1^2}}{g_1 g_2} \right], \\ V_0 &= -20m^2 \left[\frac{g_1 g_2}{3m \sqrt{g_2^2 - g_1^2}} \right]^{\frac{3}{2}}. \end{aligned} \quad (2.32)$$

According to the AdS/CFT correspondence, these vacua would correspond to $N = 2$ SCFTs in five dimensions. In this paper, we are interested in supersymmetric solutions of the form $AdS_4 \times \Sigma$ for Σ being a topological disk with a non-trivial $U(1)$ holonomy on the boundary. The solutions are holographically dual to three-dimensional SCFTs arising from compactifications of these $N = 2$ SCFTs in five dimensions on Σ .

3 $AdS_4 \times \Sigma$ solutions with $SO(2) \times SO(2)$ symmetry

We begin with $AdS_4 \times \Sigma$ solutions preserving $SO(2) \times SO(2)$ symmetry generated by J^{12} and $\tilde{J}^{12}$. It turns out that the resulting solutions lie within the $U(1) \times U(1)$ truncation of matter-coupled $F(4)$ gauged supergravity that can in turn be embedded in massive type IIA theory. Accordingly, we will first find the solutions from the six-dimensional gauged supergravity and then uplift the solutions to ten dimensions via a consistent truncation given in [43].

3.1 The six-dimensional solution

We now look for supersymmetric $AdS_4 \times \Sigma$ solutions by solving the relevant BPS equations obtained from setting the supersymmetry transformations of fermionic fields to zero. Apart from the dilaton σ , there are two singlet scalars from the $SO(4, 3)/(SO(4) \times SO(3))$ with coset representative given by

$$L = e^{\phi_1 Y_{03}} e^{\phi_2 Y_{33}}. \quad (3.1)$$

Following [52], we take the ansatz for the six-dimensional metric to be

$$ds_6^2 = f(r) ds_{AdS_4}^2 + h_1(r) dr^2 + h_2(r) d\theta^2 \quad (3.2)$$

where the metric on AdS_4 with unit radius takes the form

$$ds_{AdS_4}^2 = \frac{1}{\rho^2} (dx_{1,2}^2 + d\rho^2) \quad (3.3)$$

with $dx_{1,2}^2 = \eta_{mn} dx^m dx^n$, $m, n = 0, 1, 2$ being the flat metric on the three-dimensional Minkowski space Mkw_3 . r and θ are respectively radial and angular coordinates on the topological disk Σ . The six-dimensional curved and flat space-time indices will be split into $\mu = (m, \rho, r, \theta)$ and $\hat{\mu} = (\hat{m}, \hat{\rho}, \hat{r}, \hat{\theta})$, respectively.

In addition, there are two $SO(2) \times SO(2)$ gauge fields A^3 and A^6 with the following ansatz

$$A^3 = A_1(r) d\theta \quad \text{and} \quad A^6 = A_2(r) d\theta. \quad (3.4)$$

The Killing spinors for unbroken supersymmetry take the form

$$\epsilon_A = n_A \vartheta \otimes \eta \quad (3.5)$$

in which n_A are two components of a constant object in the doublet representation of $SO(3)_R$. η is a two-component spinor on Σ depending on r and θ coordinates while ϑ is a four-component Killing spinor on AdS_4 satisfying

$$\nabla_{\hat{a}}^{AdS_4} \vartheta = \frac{is}{2} \gamma_* \gamma_{\hat{a}} \vartheta \quad (3.6)$$

with $\hat{a} = (\hat{m}, \hat{\rho}) = 0, 1, 2, 3$ being a flat index along AdS_4 and the constant $s = \pm 1$.

By imposing the projector of the form

$$\sigma_{AB}^3 n^B = -n_A, \quad (3.7)$$

we can derive all the relevant BPS equations. The complete analysis is presented in appendix B. However, this leads to a highly complicated form of solutions. In this section, to make further analysis more traceable, we will consider a simpler solution obtained from the general one given in appendix B by imposing a relation between two integration constants arising from solving field equations for the gauge fields. In this truncated solution, the six-dimensional metric is given by

$$ds_6^2 = \frac{Br^{\frac{1}{12}}}{4m^2(G^2r - 1)^{\frac{1}{3}}} [ds_{AdS_4}^2 + ds_\Sigma^2] \quad (3.8)$$

$$\text{with} \quad ds_\Sigma^2 = \frac{r^{-\frac{4}{3}}}{36W(G^2r - 1)^{\frac{5}{3}}} dr^2 + \frac{4C^2W}{B} d\theta^2. \quad (3.9)$$

r is a real coordinate in an interval to be determined below while θ is an angular coordinate with the period chosen to be 2π . The function $W(r)$ is defined by

$$W = B - r^{\frac{2}{3}}(G^2r - 1)^{\frac{1}{3}}. \quad (3.10)$$

The three parameters are defined as

$$B = \frac{2^{\frac{4}{3}} b^{\frac{2}{3}} m^2}{3^{\frac{2}{3}} s^{\frac{2}{3}}}, \quad G = \frac{g_1}{3m}, \quad C = \frac{3}{2} g_1 m^2 C \quad (3.11)$$

where b and C are two real integration constants.

It turns out that consistency of the BPS equations and the field equations requires $\phi_1 = 0$. The solutions for the remaining two scalar fields σ and ϕ_2 are given by

$$\sigma = \frac{1}{8} \ln r \quad \text{and} \quad \phi_2 = \ln \left(\sqrt{G^2r - 1} + G\sqrt{r} \right). \quad (3.12)$$

The two $SO(2) \times SO(2)$ gauge fields have non-vanishing components only along the θ direction and take the form

$$A_\theta^3 = \frac{2}{3Gm} [C(3G^2r - 2) - q] \quad \text{and} \quad A_\theta^6 = \frac{2C}{m} \sqrt{r(G^2r - 1)} + c \quad (3.13)$$

in which q and c are constants. With the two-form field $B_{\mu\nu}$ set to zero, the corresponding non-vanishing components of the gauge field strength tensors are given by

$$\hat{F}_{r\theta}^\Lambda = F_{r\theta}^\Lambda = \frac{GC}{m} \delta_3^\Lambda + \frac{C(2G^2r - 1)}{2m\sqrt{r(G^2r - 1)}} \delta_6^\Lambda. \quad (3.14)$$

At this point, we observe that the angular coordinate θ always enters the six-dimensional metric and the field strength $F_{r\theta}$ in the combination $Cd\theta$.

We now consider the range of r for obtaining regular solutions with all the metric functions positive definite and all scalar fields real. We find that regular solutions are possible only in the following range

$$\frac{1}{G^2} < r < r_1 \quad (3.15)$$

for $B > 0$, $G > 0$, and $m > 0$ where r_1 is the real root of $W(r_1) = 0$ given by

$$r_1 = \frac{1}{3G^2}(1 + X + X^{-1}) \quad (3.16)$$

for

$$X = \left[x + \sqrt{x(x+2)} + 1 \right]^{\frac{1}{3}} \quad (3.17)$$

with

$$x = \frac{27}{2}B^3G^4. \quad (3.18)$$

We note that there are also two complex roots of the equation $W(r) = 0$. In terms of X defined in (3.17), these are given by

$$r_{c\pm} = \frac{(X-1)}{6G^2X} \left[(1-X) \pm i\sqrt{3}(X+1) \right]. \quad (3.19)$$

Since X is a non-negative real number, both of these roots, corresponding to the upper and lower signs, are complex and are not relevant to the present analysis.

For an illustrative purpose, we plot representative solutions of $f(r)$, $h_1(r)$, and $h_2(r)$ with $B = G = \mathcal{C} = 1$ and $m = \frac{1}{2}$ in figure 1. As can be seen from the figure, all the warp factors in the metric (3.2) are positive in the range $\frac{1}{G^2} < r < r_1$.

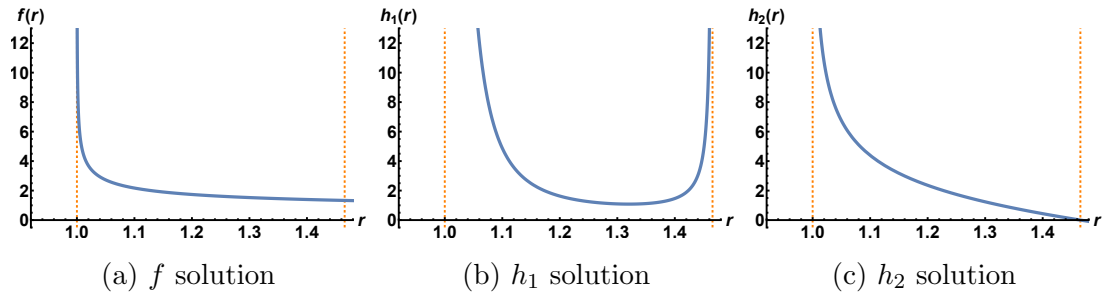


Figure 1: Numerical plots of the warp factors for $SO(2) \times SO(2)$ symmetric solution with $B = G = \mathcal{C} = 1$ and $m = \frac{1}{2}$. The solution is regular in the range $\frac{1}{G^2} = 1 < r < r_1 = 1.46$ with the two vertical dashed lines representing the two boundaries.

This result is very similar to the $SO(2) \times SO(2)$ symmetric solution found in seven-dimensional $N = 2$ gauged supergravity in [57]. As $r \rightarrow r_1$, the six-dimensional metric is approximately given by

$$ds_6^2 \approx \frac{Br_1^{\frac{1}{12}}}{4m^2(G^2r_1 - 1)^{\frac{1}{3}}} \left[ds_{AdS_4}^2 + \frac{d\rho^2 + 4\mathcal{C}^2(3G^2r_1 - 2)^2\rho^2 d\theta^2}{-9W'(r_1)r_1^{\frac{4}{3}}(G^2r_1 - 1)^{\frac{5}{3}}} \right] \quad (3.20)$$

in which ρ is the new radial coordinate defined by $\rho = \sqrt{r_1 - r}$. The θ -circle shrinks smoothly giving rise to an $\mathbb{R}^2/\mathbb{Z}_l$ orbifold at $r = r_1$ if we impose

$$\mathcal{C} = \frac{1}{2l(3G^2r_1 - 2)}, \quad l = 1, 2, 3, \dots \quad (3.21)$$

In this case, the function $(3G^2r_1 - 2)$ depends on the two constants G and B . However, its explicit form obtained from equation (3.16) is rather complicated, so we will only show that $(3G^2r_1 - 2)$ is strictly positive in the range of the r coordinate under consideration by giving a numerical plot of $(3G^2r_1 - 2)$ in figure 2.

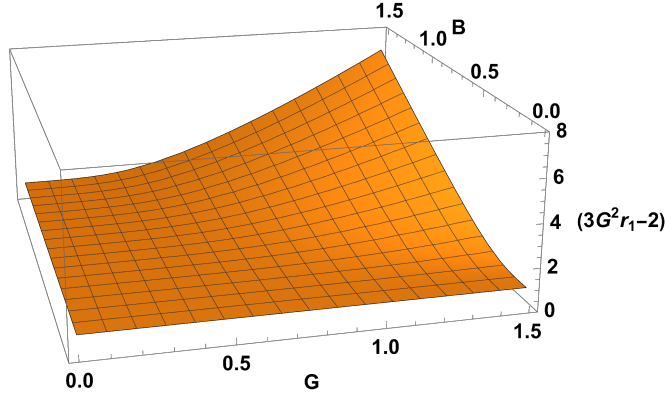


Figure 2: A numerical plot of the function $(3G^2r_1 - 2)$ in condition (3.21). Note that $(3G^2r_1 - 2) \geq 1$ for all $B > 0$ and $G > 0$.

On the other hand, as $r \rightarrow \frac{1}{G^2}$, the six-dimensional metric becomes conformal to a product of AdS_4 and a cylinder. With the new radial coordinate R given by $r = \frac{1}{G^2} + 64\mathcal{C}^6G^2B^3R^6$, the metric near $R = 0$ is given by

$$ds_6^2 \approx \frac{1}{16\mathcal{C}^2G^{\frac{3}{2}}m^2R^2} [ds_{AdS_4}^2 + 4\mathcal{C}^2(dR^2 + d\theta^2)] \quad (3.22)$$

It should be noted that in this limit the S^1 parametrized by θ decompactifies as also pointed out in [52]. In addition, by changing to another coordinate given by

$$\rho = \frac{R^2}{4\mathcal{C}^2}, \quad (3.23)$$

we find, up to an overall constant and some rescaling in $ds_{AdS_4}^2$,

$$ds_6^2 \sim \frac{1}{\rho} (ds_{AdS_4}^2 + d\theta^2) + \frac{d\rho^2}{4\rho^2}. \quad (3.24)$$

This is an AdS_6 space in Fefferman-Graham coordinate with the holographic boundary given by $AdS_4 \times S^1$. This is similar to the seven-dimensional solution studied in [68] that leads to a holographic dual of a codimension-2 defect within $N = (1, 0)$ SCFT in six dimensions. Therefore, the present solution could also be interpreted holographically as a codimension-2 defect within the five-dimensional $N = 2$ SCFT dual to the AdS_6 geometry with $SO(3) \times SO(3)$ symmetry. We also note that in the limit $r \rightarrow \frac{1}{G^2} = \frac{9m^2}{g_1^2}$, the scalar fields become

$$\sigma = \frac{1}{4} \ln \left[\frac{3m}{g_1} \right] \quad \text{and} \quad \phi_2 = 0 \quad (3.25)$$

which are precisely the values at the AdS_6 vacuum.

Topologically, the space Σ with metric (3.9) forms a disk with an origin at $r = r_1$ and a boundary at $r = \frac{1}{G^2}$. On this boundary, the θ -circle does not collapse while there is a $\mathbb{Z}_l$ orbifold singularity at the center of the disk. By using the Gauss-Bonnet theorem, we can calculate the Euler characteristic of Σ from the metric ds_Σ^2 given in (3.9)

$$\begin{aligned} \chi(\Sigma) &= \frac{1}{4\pi} \int_\Sigma R_\Sigma \text{vol}_\Sigma = \frac{2\mathcal{C}(G^2 r_1 - 1)^{\frac{1}{6}} r_1^{\frac{1}{3}} (3G^2 r_1 - 2)}{\sqrt{B}} \\ &= 2\mathcal{C} (3G^2 r_1 - 2) = \frac{1}{l} \end{aligned} \quad (3.26)$$

which corresponds to a disk with an $\mathbb{R}^2/\mathbb{Z}_l$ orbifold singularity at $r = r_1$. Noted that the integration has been performed on the interval $\frac{1}{G^2} < r < r_1$ and $0 < \theta < 2\pi$, and we have used $B = r_1^{\frac{2}{3}} (G^2 r_1 - 1)^{\frac{1}{3}}$ as obtained from $W(r_1) = 0$.

In order for the $SO(2) \times SO(2)$ gauge fields to vanish at $r = r_1$, we fix the constants q and c_2 to be

$$q = \mathcal{C} (3G^2 r_1 - 2) \quad \text{and} \quad c_2 = -\frac{2\mathcal{C}}{m} \sqrt{r_1 (G^2 r_1 - 1)} \quad (3.27)$$

leading to

$$\begin{aligned} A_\theta^3 &= \frac{2\mathcal{C}G}{m} (r - r_1) \\ \text{and} \quad A_\theta^6 &= \frac{2\mathcal{C}}{m} \left(\sqrt{r(G^2 r - 1)} - \sqrt{r_1(G^2 r_1 - 1)} \right). \end{aligned} \quad (3.28)$$

We also note that A_θ^3 is well-defined for all values of r while A_θ^6 is complex for $r < \frac{1}{G^2}$.

Finally, we give the explicit form of the Killing spinors corresponding to the unbroken supersymmetry. As previously mentioned, the solution preserves $\frac{1}{2}$ of the original supersymmetry due to the projector (B.50). For the coordinate r in the range $\frac{1}{G^2} < r < r_1$, the two-component Killing spinor $\hat{\eta}(r)$ takes the form of

$$\hat{\eta}(r) = \frac{Y_0 r^{\frac{1}{48}}}{(G^2 r - 1)^{\frac{1}{12}}} \begin{pmatrix} \sqrt{\sqrt{B} - s [r^2 (G^2 r - 1)]^{\frac{1}{6}}} \\ \sqrt{\sqrt{B} + s [r^2 (G^2 r - 1)]^{\frac{1}{6}}} \end{pmatrix} \quad (3.29)$$

where Y_0 is a real constant. Since $[r_1^2 (G^2 r_1 - 1)]^{\frac{1}{3}} = B$, we find, as $r \rightarrow r_1$, that

$$\hat{\eta}(r_1) = Y_0 r_1^{3/16} \begin{pmatrix} \sqrt{1-s} \\ \sqrt{1+s} \end{pmatrix}. \quad (3.30)$$

Recall that the parameter $s = \pm 1$, we readily see that only one of the two components is non-vanishing. Therefore, only $\frac{1}{4}$ of the original supersymmetry is preserved at the origin of the disk at $r = r_1$. Moreover, $\hat{\eta}$ is well-behaved near $r = r_1$ and hence globally defined on the disk Σ .

3.2 Uplift to massive type IIA theory

As previously mentioned, the resulting solution lies within the $U(1) \times U(1)$ sector of the matter-coupled $F(4)$ gauged supergravity which can be embedded in massive type IIA theory by a consistent truncation on a half four-sphere. The corresponding truncation ansatz is reviewed in appendix E. Using the formulae in appendix E, we find the ten-dimensional metric in string frame of the form

$$\begin{aligned} ds_{10}^2 = & \cos^{-\frac{1}{3}} \xi e^{-\frac{\sigma}{2}} \Delta^{\frac{1}{2}} \left\{ \frac{Br^{\frac{1}{12}}}{4m^2(r-1)^{\frac{1}{3}}} ds_{AdS_4}^2 + \frac{Br^{-\frac{5}{4}}}{36W(r-1)^{\frac{5}{3}}} dr^2 + \frac{CW r^{\frac{1}{12}}}{m^2(r-1)^{\frac{1}{3}}} \right. \\ & + \frac{\Delta^{-1} e^{-\sigma}}{9m^2} [\sin^2 \xi (e^{-\phi_2} \cos^2 \eta + e^{\phi_2} \sin^2 \eta) d\eta^2 - \sin 2\xi \sin 2\eta \sinh \phi_2 d\xi d\eta \\ & + [e^{4\sigma} \sin^2 \xi + \cos^2 \xi (e^{-\phi_2} \sin^2 \eta + e^{\phi_2} \cos^2 \eta)] d\xi^2 \\ & \left. + \sin^2 \xi [\cos^2 \eta e^{\phi_2} (d\varphi_2 - g_1 \mathcal{A}_2)^2 + \sin^2 \eta e^{-\phi_2} (d\varphi_1 - g_1 \mathcal{A}_1)^2] \right\} \quad (3.31) \end{aligned}$$

in which for convenience, we have set $g_1 = 3m$ leading to $G = 1$. The gauge fields $\mathcal{A}_1$ and $\mathcal{A}_2$ are given by

$$\begin{aligned} \mathcal{A}_1 &= \frac{\mathcal{C}}{m} \left\{ r - r_1 + \sqrt{r(r-1)} - \sqrt{r_1(r_1-1)} \right\} \\ \text{and} \quad \mathcal{A}_2 &= \frac{\mathcal{C}}{m} \left\{ r - r_1 - \sqrt{r(r-1)} + \sqrt{r_1(r_1-1)} \right\}. \quad (3.32) \end{aligned}$$

We can also use the formulae given in appendix E to compute the holographic free energy of the dual SCFT in three dimensions. However, the solution

leads to an infinite free energy due to the decompactified θ -circle at $r = 1$. The infinity arises near the AdS_6 asymptotic geometry as pointed out in [43]. Following [68], the solution could be holographically interpreted as the presence of a three-dimensional conformal defect within the $N = 2$ SCFT in five dimensions. It would be interesting to further study the solution along this line and perform some holographic computation as in [68] and [69].

4 $AdS_4 \times \Sigma$ solutions with $SO(2)_{\text{diag}}$ symmetry

In this section, we repeat the same analysis within a truncation to $SO(2)_{\text{diag}}$ invariant sector. There are four singlet scalars from $SO(4, 3)/(SO(4) \times SO(3))$ coset under $SO(2)_{\text{diag}}$ symmetry generated by $J^{12} + \tilde{J}^{12}$. These singlets can be described by the coset representative

$$L = e^{\phi_0 Y_{03}} e^{\phi_1 (Y_{11} + Y_{22})} e^{\phi_2 Y_{33}} e^{\phi_3 (Y_{12} - Y_{21})} . \quad (4.1)$$

To implement the $SO(2)_{\text{diag}}$ symmetry, we impose the following constraint on the two $SO(2) \times SO(2)$ gauge fields

$$g_1 A^3 = -g_2 A^6 . \quad (4.2)$$

The detailed analysis of the BPS equations is shown in appendix C. It turns out that consistency requires all but ϕ_2 to vanish.

In this case, the six-dimensional metric reads

$$ds_6^2 = \frac{Br^{\frac{5}{12}}}{4m^2 (s(G^2 r - 1))^{\frac{2}{3}}} [ds_{AdS_4}^2 + ds_\Sigma^2] \quad (4.3)$$

$$\text{with} \quad ds_\Sigma^2 = \frac{r^{-\frac{5}{3}}}{9W (s(G^2 r - 1))^{\frac{4}{3}}} dr^2 + \frac{4\mathcal{C}^2 W}{B} d\theta^2 \quad (4.4)$$

where the function $W(r)$ is now given by

$$W = B - r^{\frac{1}{3}} (s(G^2 r - 1))^{\frac{2}{3}} . \quad (4.5)$$

The three parameters appearing in the above equations are defined by

$$B = \left(\frac{8bg_1 m^2}{9} \right)^{\frac{2}{3}}, \quad G = \frac{g_1}{3m}, \quad \mathcal{C} = 3g_1 m^2 C \quad (4.6)$$

where b and C are real integration constants.

The solutions for the dilaton σ and the only non-vanishing scalar from the vector multiplets ϕ_2 are given by

$$\sigma = \frac{1}{8} \ln r \quad \text{and} \quad \phi_2 = \frac{1}{2} \ln r + \ln G . \quad (4.7)$$

At this point, we note that the analysis performed in appendix C also leads to a relation between the two $SO(3)$ coupling constants namely $g_2 = -g_1$. However, when ϕ_1 and ϕ_3 vanish, the coupling constant g_2 no longer appears in the solutions. Accordingly, the resulting solutions still lie within the $U(1) \times U(1)$ sector of the matter-coupled $F(4)$ gauged supergravity and can be uplifted to massive type IIA theory as in the previous section.

In this case, we still have $B_{\mu\nu} = 0$, and the two gauge fields A^3 and A^6 are equal with non-vanishing components given by

$$A_\theta^3 = A_\theta^6 = \frac{\mathcal{C}(3G^2r - 1) - 2q}{3Gm}. \quad (4.8)$$

The two-component Killing spinor takes the form of

$$\hat{\eta}(r) = \frac{Y_0 r^{\frac{5}{48}}}{(s(G^2r - 1))^{\frac{1}{6}}} \begin{pmatrix} \sqrt{\sqrt{B} - sr^{\frac{1}{6}}(s(G^2r - 1))^{\frac{1}{3}}} \\ \sqrt{\sqrt{B} + sr^{\frac{1}{6}}(s(G^2r - 1))^{\frac{1}{3}}} \end{pmatrix} \quad (4.9)$$

with Y_0 being a real constant.

We now determine the range of the coordinate r leading to regular solutions. Unlike in the previous section, for suitable values of the parameters B and G , there can be up to three roots for $W(r) = 0$ equation. For $(27B^3G^2 - 4) > 0$ or $B > \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$, there is only one root given by

$$r_0 = \frac{1}{3} \left[\frac{2}{G^2} + \frac{1}{\tilde{Z}} + \frac{\tilde{Z}}{G^4} \right] \quad (4.10)$$

with

$$\tilde{Z} = \left[\frac{27B^3G^8}{2} - G^6 + \frac{3}{2}\sqrt{3}\sqrt{B^3G^{14}(27B^3G^2 - 4)} \right]^{\frac{1}{3}}. \quad (4.11)$$

On the other hand, for $(27B^3G^2 - 4) < 0$ or $B < \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$, there are three roots given by

$$r_0 = \frac{1}{3} \left[\frac{2}{G^2} + \frac{1}{Z} + \frac{1}{Z^*} \right] \quad (4.12)$$

$$\text{and} \quad r_{\pm} = \frac{1}{6} \left[\frac{4}{G^2} - \frac{1}{Z} - \frac{1}{Z^*} \pm \sqrt{3}i \left(\frac{1}{Z^*} - \frac{1}{Z} \right) \right] \quad (4.13)$$

for

$$Z = \left[\frac{27B^3G^8}{2} - G^6 + i\frac{3\sqrt{3}}{2}\sqrt{B^3G^{14}(4 - 27B^3G^2)} \right]^{\frac{1}{3}}. \quad (4.14)$$

We also note that for $B = \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$ the two roots r_+ and r_- degenerate. Furthermore, the behaviors of the solution for $s = +1$ and $s = -1$ are different, so we will discuss each case separately.

4.1 Solutions with $s = +1$

For the case of $s = +1$, we find a regular solution in the range

$$\frac{1}{G^2} < r < r_0 \quad (4.15)$$

for $B > 0$, $G > 0$, and $m > 0$. An example of numerical plots for the three warp factors with $s = 1$, $B = G = \mathcal{C} = 1$, and $m = \frac{1}{2}$ is shown in figure 3. This is very similar to the previous case with $SO(2) \times SO(2)$ symmetry. As $r \rightarrow \frac{1}{G^2}$, the six-dimensional metric is conformal to a product of AdS_4 and a cylinder with the same metric given in (3.22) but with the new radial coordinate R given by $r = \frac{1}{G^2} + \frac{8\mathcal{C}^3 B^{\frac{3}{2}} R^3}{G}$. As in the previous section, the S^1 with the coordinate θ decompactifies in this limit leading to a locally asymptotic AdS_6 geometry with a holographic boundary given by $AdS_4 \times S^1$. In this limit, the scalar fields become

$$\sigma = \frac{1}{4} \ln \left[\frac{3m}{g_1} \right] \quad \text{and} \quad \phi_2 = 0 \quad (4.16)$$

as expected at the AdS_6 vacuum with $SO(3) \times SO(3)$ symmetry.

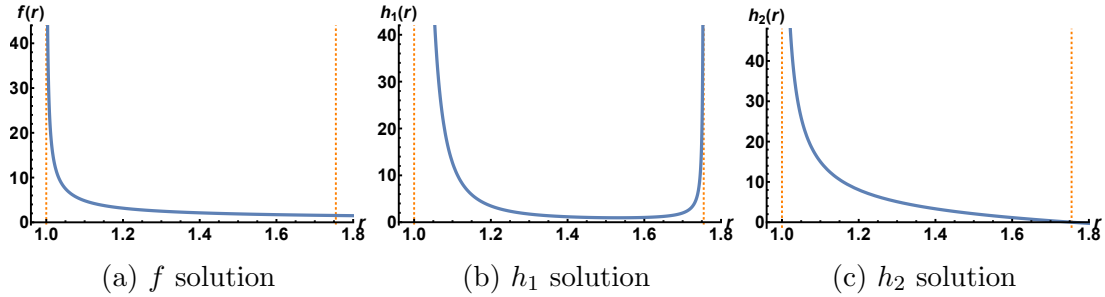


Figure 3: Numerical plots of the warp factors for the $SO(2)_{\text{diag}}$ symmetric solution with $s = 1$, $B = G = \mathcal{C} = 1$, and $m = \frac{1}{2}$. The solution is regular in the range $\frac{1}{G^2} = 1 < r < r_0 = 1.755$ with the two vertical dashed lines representing the two boundaries.

On the other hand, as $r \rightarrow r_0$, we find the six-dimensional metric of the form

$$ds_6^2 \approx \frac{Br_0^{\frac{5}{12}}}{4m^2 (G^2 r_0 - 1)^{\frac{2}{3}}} \left[ds_{AdS_4}^2 + \frac{4 \left[d\rho^2 + \mathcal{C}^2 (3G^2 r_0 - 1)^2 \rho^2 d\theta^2 \right]}{\left[-9W'(r_0) r_0^{\frac{5}{3}} (G^2 r_0 - 1)^{\frac{4}{3}} \right]} \right] \quad (4.17)$$

after changing to the new radial coordinate $\rho = \sqrt{r_0 - r}$. In this case, the θ -circle shrinks smoothly near the endpoint at $r = r_0$ if we choose

$$\mathcal{C} = \frac{1}{l(3G^2 r_0 - 1)}, \quad l = 1, 2, 3, \dots \quad (4.18)$$

Since the explicit form of $(3G^2r_0 - 1)$ derived from (4.12) is complicated, we only numerically verify that this function is always greater than 2 within the regularity range as shown in figure 4.

As in the previous case, we can compute the Euler characteristic of Σ from the metric (4.4). The result is given by

$$\begin{aligned}\chi(\Sigma) &= \frac{1}{4\pi} \int_{\Sigma} R_{\Sigma} \text{vol}_{\Sigma} = \frac{\mathcal{C} r_0^{1/6} (G^2 r_0 - 1)^{1/3} (3G^2 r_0 - 1)}{\sqrt{B}} \\ &= \mathcal{C} (3G^2 r_0 - 1) = \frac{1}{l}\end{aligned}\tag{4.19}$$

which again shows that Σ is a disk with an $\mathbb{R}^2/\mathbb{Z}_l$ orbifold singularity at $r = r_0$.

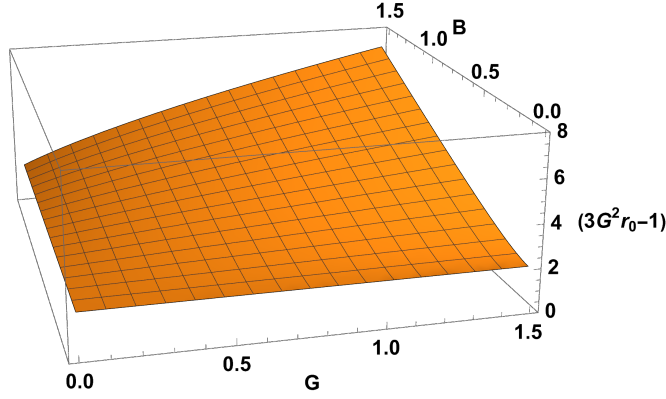


Figure 4: A numerical plot of the function $(3G^2r_0 - 1)$ under the condition (4.18). Note that $(3G^2r_0 - 1) \geq 2$ for all $B > 0$ and $G > 0$.

To obtain the $SO(2)_{\text{diag}}$ gauge field that vanishes at $r = r_0$, we choose the constant q to be

$$q = \frac{\mathcal{C}}{2} (3G^2r_0 - 1) = \frac{1}{2l}\tag{4.20}$$

resulting in

$$A_{\theta}^3 = A_{\theta}^6 = \frac{\mathcal{C}G}{m} (r - r_0).\tag{4.21}$$

Finally, the Killing spinor $\hat{\eta}$ at the end point $r = r_0$ is given by

$$\hat{\eta}(r_1) = \sqrt{2} Y_0 r_0^{\frac{3}{16}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.\tag{4.22}$$

This implies that the solution is $\frac{1}{4}$ -BPS at the orbifold singularity.

4.2 Solutions with $s = -1$ and $0 < B < \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$

A more interesting case occurs when $s = -1$ and $0 < B < \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$. With $G > 0$ and $m > 0$, we find regular solutions in the following two ranges

$$i) \quad 0 < r < r_+, \quad (4.23)$$

$$ii) \quad r_- < r < \frac{1}{G^2}. \quad (4.24)$$

An example of numerical plots of the three warp factors for $s = -1$, $B = 0.524$, $G = \mathcal{C} = 1$, and $m = \frac{1}{2}$ is shown in figure 5.

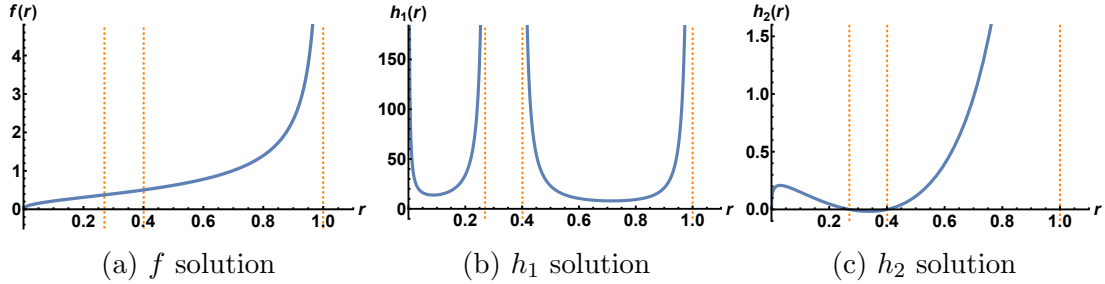


Figure 5: Numerical plots of the warp factors for the $SO(2)_{\text{diag}}$ symmetric solution evaluated at $s = -1$, $B = 0.524$, $G = \mathcal{C} = 1$, and $m = \frac{1}{2}$. The solution is regular within the intervals $0 < r < r_+ = 0.27$ and $r_- = 0.4 < r < \frac{1}{G^2} = 1$. The three vertical dashed lines mark the boundaries of these regions (excluding $r = 0$).

As $r \rightarrow \frac{1}{G^2}$, we can change to a new radial coordinate R given by $r = \frac{1}{G^2} - \frac{8\mathcal{C}^3 B^{\frac{3}{2}} R^3}{G}$ and obtain the six-dimensional metric given in (3.22). This again leads to the AdS_6 asymptotic geometry with an $AdS_4 \times S^1$ boundary. On the other hand, as $r \rightarrow 0$, the metric near $r = 0$ reads

$$ds_6^2 \approx \frac{B^{\frac{9}{4}} \mathcal{C}^{\frac{5}{2}} R^{\frac{5}{2}}}{4m^2} [ds_{AdS_4}^2 + 4\mathcal{C}^2 (dR^2 + d\theta^2)] \quad (4.25)$$

with the new radial coordinate R given by $r = B^3 \mathcal{C}^6 R^6$. In this case, the six-dimensional metric becomes conformal to a product of AdS_4 and a cylinder with the warp factor of AdS_4 vanishing. This potentially leads to a curvature singularity. We will see later that the uplifted ten-dimensional solution is not singular.

As $r \rightarrow r_{\pm}$, we can introduce a new radial coordinate $\rho = \sqrt{r_{\pm} - r}$ and find the six-dimensional metric, as $r \rightarrow r_{\pm}$ or $\rho \rightarrow 0$, of the form

$$ds_6^2 \approx \frac{Br_{\pm}^{\frac{5}{12}}}{4m^2 (1 - G^2 r_{\pm})^{\frac{2}{3}}} \left[ds_{AdS_4}^2 + \frac{4 \left[d\rho^2 + \mathcal{C}^2 (1 - 3G^2 r_{\pm})^2 \rho^2 d\theta^2 \right]}{\left[-9W'(r_{\pm}) r_{\pm}^{\frac{5}{3}} (1 - G^2 r_{\pm})^{\frac{4}{3}} \right]} \right]. \quad (4.26)$$

In this case, the θ -circle shrinks smoothly at $r = r_{\pm}$ if we impose

$$\mathcal{C} = \frac{1}{l|1 - 3G^2r_{\pm}|}, \quad l = 1, 2, 3, \dots \quad (4.27)$$

leading to an $\mathbb{R}^2/\mathbb{Z}_l$ orbifold as before. We have used an absolute sign for the function $(1 - 3G^2r_{\pm})$ to ensure that the constant $\mathcal{C}$ is positive. In particular, using $r_{\pm}$ given in (4.13), we find that $0 < (1 - 3G^2r_-) < 2$ and $-1 < (1 - 3G^2r_+) < 0$. This can also be seen from a numerical plot shown in figure 6.

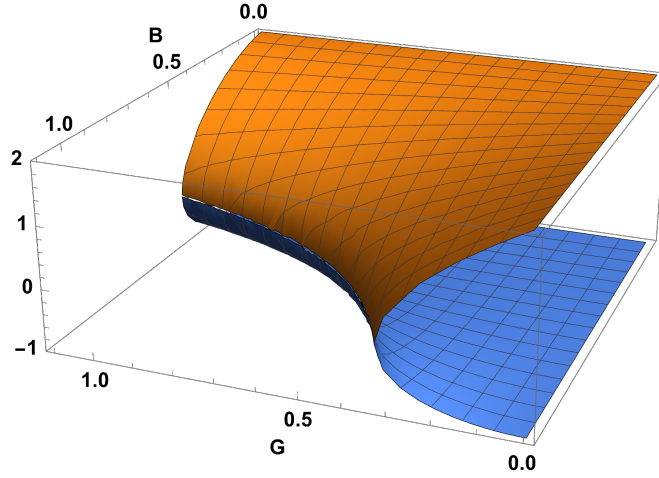


Figure 6: Numerical plot of the functions $(1 - 3G^2r_-)$ (orange surface) and $(1 - 3G^2r_+)$ (blue surface) in condition (4.27). Both functions approach zero as $B \rightarrow \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$.

With the condition (4.27), the Euler characteristic of Σ can be determined to be the same as in (3.26) and (4.19). Accordingly, the space Σ with the metric (4.4) forms a topological disk for both $r \in (0, r_+)$ and $r \in (r_-, \frac{1}{G^2})$ with a $\mathbb{Z}_l$ orbifold singularity at the center $r = r_{\pm}$. In order for the $SO(2)_{\text{diag}}$ gauge field to vanish at $r = r_{\pm}$, we fix the constant q such that

$$q = \frac{1}{2}\mathcal{C}(3G^2r_{\pm} - 1) \quad \text{and} \quad A_{\theta}^3 = A_{\theta}^6 = \frac{\mathcal{C}G}{m}(r - r_{\pm}) \quad (4.28)$$

which gives rise to the Killing spinor $\hat{\eta}$ at $r = r_{\pm}$ of the form

$$\hat{\eta}(r_{\pm}) = \sqrt{2}Y_0r_{\pm}^{\frac{3}{16}} \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (4.29)$$

4.3 Solutions with $s = -1$ and $B = \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$

In this case, setting $s = -1$ and $B = \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$, we find that $r_+ = r_- = r_*$. However, it turns out that the θ -circle cannot shrink smoothly at $r = r_*$ since $(1 - 3G^2r_*) = 0$

when $B = \frac{2^{2/3}}{3G^{2/3}}$. This makes $\mathcal{C}$ diverge according to condition (4.27). Instead, at $r = r_*$, the six-dimensional metric is asymptotically given by

$$ds_6^2 \approx \frac{1}{4 \times 3^{3/4} G^{3/2} m^2} \left[ds_{AdS_4}^2 + \frac{d\rho^2 + 9\mathcal{C}^2 G^4 d\theta^2}{\rho^2} \right] \quad (4.30)$$

with the new radial coordinate ρ defined by $\rho = \frac{1}{3G^2} - r$. An example of numerical plots of the three warp factors for $s = -1$, $B = \frac{2^{2/3}}{3G^{2/3}} = 0.529$, $G = \mathcal{C} = 1$, and $m = \frac{1}{2}$ is given in figure 7.

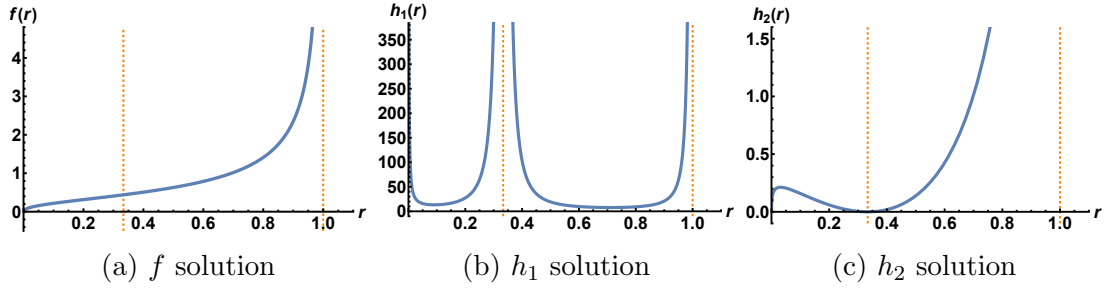


Figure 7: Numerical plots of the warp factors for the $SO(2)_{\text{diag}}$ symmetric solution evaluated at $s = -1$, $B = 0.529$, $G = \mathcal{C} = 1$, and $m = \frac{1}{2}$. The solution is regular within the intervals $0 < r < r_* = 0.33$ and $r_* < r < \frac{1}{G^2} = 1$. The two vertical dashed lines mark the boundaries of these regions (excluding $r = 0$).

4.4 Solutions with $s = -1$ and $B > \frac{2^{2/3}}{3G^{2/3}}$

Finally, for $s = -1$ and $B > \frac{2^{2/3}}{3G^{2/3}}$, there is only one root to $W(r) = 0$ equation, and we find a regular solution only in the range

$$0 < r < \frac{1}{G^2} \quad (4.31)$$

for $G > 0$ and $m > 0$. As in the previous case, the θ -circle does not shrink, and the solution interpolates between the two end points that are conformal to a product of AdS_4 and a cylinder at $r = 0$ and $r = \frac{1}{G^2}$ considered above. An example of numerical plots of the warp factors with $s = -1$, $B = G = \mathcal{C} = 1$, and $m = \frac{1}{2}$ is shown in figure 8.

4.5 Uplift to massive type IIA theory

We now consider uplifting the six-dimensional solutions to ten-dimensional massive type IIA theory. As in the previous section, at $r = \frac{1}{G^2}$, the θ -circle decompactifies resulting in an AdS_6 asymptotic geometry. This again leads to infinite

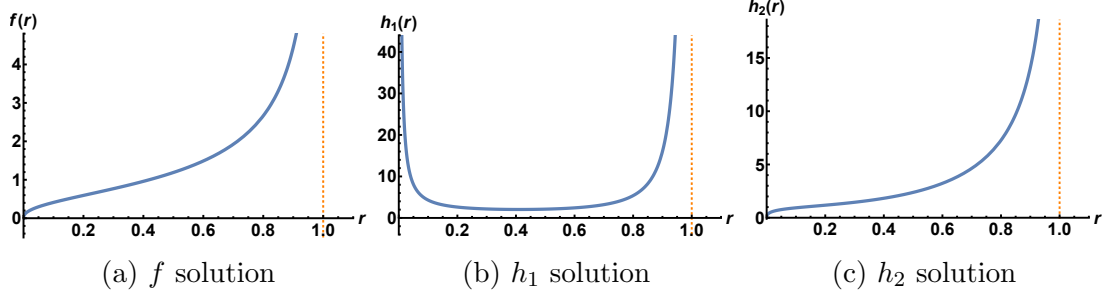


Figure 8: Numerical plots of the warp factors for the $SO(2)_{\text{diag}}$ symmetric solution evaluated at $s = -1$, $B = G = \mathcal{C} = 1$, and $m = \frac{1}{2}$. The solution is regular within the interval $0 < r < \frac{1}{G^2} = 1$. The vertical dashed line marks the boundary of the region at $r = 1$.

holographic free energy. Accordingly, we will only consider the uplifted solution in the case of $0 < r < r_+$ given in (4.23).

In the present case, we have

$$\mathcal{A}_1 = \frac{\mathcal{C}G}{m}(r - r_+) \quad \text{and} \quad \mathcal{A}_2 = 0. \quad (4.32)$$

The ten-dimensional metric is then given by

$$\begin{aligned} ds_{10}^2 = & \frac{9\lambda^2 \cos^{-\frac{1}{3}} \xi e^{-\frac{\sigma}{2}} \Delta^{\frac{1}{2}} B r^{\frac{5}{12}}}{4g_1^2(r-1)^{\frac{2}{3}}} \left\{ ds_{AdS_4}^2 + \frac{r^{-\frac{5}{3}}}{9W(r-1)^{\frac{4}{3}}} dr^2 + \frac{4CW}{B} d\theta^2 \right. \\ & + \frac{4(r-1)^{\frac{2}{3}} e^{-\frac{\sigma}{2}} \Delta^{-1}}{9Br^{\frac{5}{12}}} \left[\sin^2 \xi (e^{-\phi_2} \cos^2 \eta + e^{\phi_2} \sin^2 \eta) d\eta^2 - \sin 2\xi \sin 2\eta \times \right. \\ & \left. \sinh \phi_2 d\eta d\xi + [e^{4\sigma} \sin^2 \xi + \cos^2 \xi (e^{-\phi_2} \sin^2 \eta + e^{\phi_2} \cos^2 \eta)] d\xi^2 \right. \\ & \left. \left. + \sin^2 \xi (e^{\phi_2} \cos^2 \eta d\varphi_2^2 + e^{-\phi_2} \sin^2 \eta (d\varphi_1 - g_1 \mathcal{A}_1)^2) \right] \right\}. \quad (4.33) \end{aligned}$$

In this equation, we have set $m = \frac{g_1}{3}$ such that the parameter $G = 1$ for simplicity. The warp factor is given by

$$\Delta = e^{-3\sigma} [\cos^2 \xi + e^{4\sigma - \phi_2} \sin^2 \xi (\cos^2 \eta + e^{2\phi_2} \sin^2 \eta)]. \quad (4.34)$$

The ten-dimensional dilaton takes the form

$$e^\Phi = \lambda^2 \cos^{-\frac{5}{6}} \xi \Delta^{\frac{1}{4}} e^{-\frac{5\sigma}{4}}. \quad (4.35)$$

The four-form field strength reads

$$\begin{aligned}
F_{(4)} = & \frac{\lambda \cos^{-\frac{2}{3}} \xi}{2g_1^3 \Delta} \left\{ U \Delta^{-1} \sin^3 \xi \cos \xi \sin 2\eta d\eta \wedge d\xi \wedge D\varphi_1 \wedge d\varphi_2 \right. \\
& + \frac{\phi'_2 e^{-6\sigma}}{\Delta} \sin 2\eta \sin^3 \xi \cos \xi \left[e^{-2\phi_2} \sin 2\xi (e^{4\phi_2} \sin^2 \eta - \cos^2 \eta) dr \wedge d\eta \right. \\
& \left. - \sin 2\eta (2 \cos^2 \xi \cosh 2\phi_2 + e^{8\sigma} \sin^2 \xi) dr \wedge d\xi \right] \wedge D\varphi_1 \wedge d\varphi_2 \\
& + \frac{1}{4} g_1 e^{-3\sigma} \sin 2\xi \mathcal{F}_1 \wedge \left[4e^{-\phi_2} \cos^2 \eta (e^{\phi_2} \cos^2 \xi + e^{4\sigma} \sin^2 \xi) d\xi \right. \\
& \left. \left. - \sin 2\eta \sin 2\xi d\eta \right] \wedge d\varphi_2 \right\}
\end{aligned} \tag{4.36}$$

with $D\varphi_1 = d\varphi_1 - g_1 \mathcal{A}_1$ and $\phi'_2 = \frac{d\phi_2}{dr}$. In all of these results, we have introduced a real parameter λ in order to properly quantize all the fluxes along the internal space, see more detail in [50]. The function U is given by

$$\begin{aligned}
U = & \frac{2}{3} e^{-6\sigma} \cos^2 \xi - \frac{2}{3} e^{-2\sigma} \left[2 \cosh \phi_2 (2 + \cos 2\xi) \right. \\
& \left. + \sin^2 \xi (3e^{4\sigma} - 2 \cos 2\eta \sinh \phi_2) \right].
\end{aligned} \tag{4.37}$$

We also recall that the solutions for σ and ϕ_2 are give in (4.7). The remaining ten-dimensional field is given by the Romans mass

$$F_{(0)} = \frac{m}{\lambda^3} = \frac{g_1}{3\lambda^3}. \tag{4.38}$$

Using the formula in appendix E, we can compute the holographic free-energy of the dual SCFT in three dimensions. The result is given by

$$\begin{aligned}
\mathbf{F} &= \frac{9\lambda^4}{80\pi^2 g_1^3 \ell_s^8} \int_0^{r_+} \frac{B^{\frac{3}{2}} \mathcal{C}}{24m^4 (r-1)^2} dr \\
&= \left(\frac{3}{2}\right)^5 \frac{\lambda^4 B^{\frac{3}{2}} \mathcal{C} r_+}{20\pi^2 g_1^7 \ell_s^8 (1-r_+)}.
\end{aligned} \tag{4.39}$$

To determine the parameter λ , we follow [50] and consider the flux quantization of $F_{(0)}$ and $F_{(4)}$ of the form

$$2\pi \ell_s F_{(0)} = n_0 = 8 - N_f \quad \text{and} \quad \frac{1}{(2\pi \ell_s)^3} \int F_{(4)} = N \tag{4.40}$$

for intergers n_0 , N and N_f . We also recall that N and N_f correspond to the numbers of D4-branes and D8-branes in the configuration, respectively. The first condition immediately gives

$$n_0 = \frac{2\pi \ell_s g_1}{3\lambda^3}. \tag{4.41}$$

By considering the integral along the half S^4 directions, we find

$$\begin{aligned} \frac{1}{(2\pi\ell_s)^3} \int F_{\eta\xi\varphi_1\varphi_2} d\eta \wedge d\xi \wedge d\varphi_1 \wedge d\varphi_2 &= \frac{\lambda(2\pi)^2}{(2\pi\ell_s)^3} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \frac{U \cos^{\frac{1}{3}} \xi \sin^3 \xi \sin 2\eta}{2g_1^3 \Delta^2} d\eta d\xi \\ &= -\frac{3\lambda}{8\pi g_1^3 \ell_s^3} \end{aligned} \quad (4.42)$$

which gives

$$N = \frac{3\lambda}{8\pi g_1^3 \ell_s^3}. \quad (4.43)$$

For consistency, we have identified the flux of $F_{(4)}$ as minus the number of D4-branes N as in [50]. The two conditions in (4.41) and (4.43) then give rise to

$$\lambda^8 = \frac{\pi^2}{9n_0 N} \quad \text{and} \quad g_1^8 = \frac{9\pi^6}{(2\pi\ell_s)^8 N^3 n_0}. \quad (4.44)$$

With all these, we can write the free energy as

$$\mathbf{F} = \frac{18\pi g_1 r_+^{\frac{3}{2}} N^2 \sqrt{N(8 - N_f)}}{5(3r_+ - 1)l} \quad (4.45)$$

in which we have used $B^{\frac{3}{2}} = r_+^{\frac{1}{2}}(r_+ - 1)$ and $\mathcal{C} = \frac{1}{(3r_+ - 1)l}$.

As previously mentioned, the six-dimensional metric when $r \rightarrow 0$ is singular as given in (4.25). We first note that the metric in this limit takes the same form as that considered in [43] upon changing the coordinates in [43] as

$$z = \frac{2\mathcal{C}}{m}\theta \quad \text{and} \quad r = m\mathcal{C}\sqrt{q_1}R \quad (4.46)$$

with

$$q_1 = \frac{B^{\frac{3}{2}}}{2^{\frac{4}{3}}m^2}. \quad (4.47)$$

As shown in [43], the six-dimensional metric with this asymptotic behavior describes topological disks with enhanced symmetry. This is one of the reasons that we identify the solution in the range $(0, r_+)$ as one example of disk solutions with enhanced symmetry, see more detail in appendix D.

Following [43], we will take the constrained coordinates μ_a to be

$$\mu_0 = \sqrt{1 - \mu^2} \cos \vartheta, \quad \mu_1 = \mu, \quad \mu_2 = \sqrt{1 - \mu^2} \sin \vartheta. \quad (4.48)$$

As $r \rightarrow 0$, the ten-dimensional metric is given by

$$\begin{aligned} ds_{10}^2 &= \frac{B^{\frac{3}{4}}}{2^{\frac{2}{3}}m^{\frac{3}{2}}} \cos^{-\frac{1}{3}} \vartheta (1 - \mu^2)^{\frac{1}{3}} \left\{ \tilde{r}^{\frac{1}{3}} (ds_{AdS_4}^2 + \mathcal{C}^2 d\theta^2) \right. \\ &\quad \left. + \frac{2^{\frac{8}{3}}m^6}{g_1^2 B^3 \tilde{r}} \left[d\tilde{r}^2 + \tilde{r}^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi_2^2) + \frac{B^{\frac{3}{2}}}{2^{\frac{4}{3}}m^3} \frac{d\mu^2 + \mu^2 d\varphi_1^2}{1 - \mu^2} \right] \right\} \end{aligned} \quad (4.49)$$

with $\tilde{r} = r^3$. As anticipated in [43], this metric describes the near-horizon limit of D4-branes within the D4-D8-system smeared along two directions. In particular, the ten-dimensional metric corresponding to this smeared D4-branes has already been given in [70] as

$$ds_{10}^2 = \left(\frac{3}{2} Q_8 \cos \vartheta \right)^{-\frac{1}{3}} \left\{ Q_4^{-\frac{1}{2}} \sigma^{\frac{1}{3}} (-dt^2 + dx_1^2 + dx_2^2 + dx_3^2 + dx_4^2) + Q_4^{\frac{1}{2}} \sigma^{-1} [dx_5^2 + dx_6^2 + d\sigma^2 + \sigma^2 (d\vartheta^2 + \sin^2 \vartheta d\psi^2)] \right\} \quad (4.50)$$

which is of the same form as (4.49). Therefore, the singularity at $r = 0$ in the six-dimensional metric corresponds to D4-branes smeared along two directions in the uplifted ten-dimensional solution of massive type IIA theory.

5 Conclusions

We have found a number of supersymmetric $AdS_4 \times \Sigma$ solutions in which Σ is a topological disk with a non-trivial $U(1)$ holonomy on the boundary or a half-spindle from matter-coupled $F(4)$ gauged supergravity in six dimensions with $SO(3) \times SO(3)$ gauge group. The resulting solutions preserve eight supercharges and $SO(2) \times SO(2)$ or $SO(2)_{\text{diag}}$ symmetries and represent a new class of supersymmetric solutions from matter-coupled $F(4)$ gauged supergravity in six dimensions. We have also extensively discussed various possible ranges of the radial coordinate in which regular solutions exist. The results are very similar to supersymmetric $AdS_5 \times \Sigma$ solutions found in seven-dimensional $N = 2$ gauged supergravity coupled to three vector multiplets studied in [57].

In the case of $SO(2) \times SO(2)$ symmetric solutions, only a single range of the radial coordinate leads to regular solutions. It turns out that, unlike the solution from pure $F(4)$ gauged supergravity studied previously in [54], the solution is asymptotic to AdS_6 geometry at which the circle inside Σ decompactifies. This leads to infinite holographic free energy as also pointed out in [43]. The solution is expected to describe a codimension-2 conformal defect within the five-dimensional $N = 2$ SCFT similar to the solution considered in [68] and [69] in which a holographic description of codimension-2 defects within six-dimensional $N = (1, 0)$ and five-dimensional $N = 2$ SCFTs has been given.

For $SO(2)_{\text{diag}}$ symmetric solutions, there are a number of possible ranges of the radial coordinate for which regular solutions exist. Some of the solutions are similar to that of pure $F(4)$ gauged supergravity and lead to finite free energy upon uplifted to massive type IIA theory. These solutions should lead to new three-dimensional SCFTs arising from compactifications of $N = 2$ SCFT in five dimensions on a half-spindle. On the other hand, there are solutions with AdS_6 asymptotic geometry as in the $SO(2) \times SO(2)$ case. These solutions also lead to infinite free energy and could also be interpreted as gravity duals of codimension-2 defects within $N = 2$ SCFT in five dimensions. We have also found that some

of the solutions lie outside the space of solutions given in [43] and represent new solutions of this type, see more detail in appendix D. Finally, we briefly mention that if we consider solutions with $SO(2)_R$ symmetry from the matter-coupled $F(4)$ gauged supergravity, the solutions turn out to be the same as that found in [54] from pure $F(4)$ gauged supergravity. In particular, consistency of the field equations for the $SO(2)_R$ gauge field requires that all scalars from vector multiplets must vanish.

As in other cases, the $AdS_4 \times \Sigma$ solutions found here are just dual to the IR limit of the full solutions describing holographic RG flows across dimensions from five to three dimensions. It could be interesting to construct the full solutions interpolating between this $AdS_4 \times \Sigma$ geometry and the AdS_6 vacuum dual to the five-dimensional SCFT. The solutions with AdS_6 asymptotic geometry also deserve further investigation in the context of conformal defects by using holographic renormalization as performed in [68] to determine vacuum expectation values of the stress tensor and currents in the dual SCFT or entanglement entropy of the defects as in [69]. These three-dimensional defects would align nicely with line and surface defects as well as Janus interfaces studied in [66, 67, 73]. It is also interesting to find $AdS_4 \times \Sigma$ solutions from matter-coupled $F(4)$ gauged supergravity with $ISO(3)$ gauge group. This gauged supergravity has been studied recently in [74, 75, 76] and can be uplifted to type IIB theory as shown in [77]. The solutions could similarly give new three-dimensional SCFTs or codimension-2 defects within $N = 2$ SCFTs in five dimensions realized in terms of brane configurations in type IIB theory.

A Bosonic field equations

In this appendix, we collect all the bosonic field equations obtained from the Lagrangian given in (2.4). These are given by

$$0 = \frac{3}{8}D_\rho(e^{4\sigma}H^{\mu\nu\rho}) - m^2e^{-2\sigma}\mathcal{N}_{00}B^{\mu\nu} + me^{-2\sigma}\mathcal{N}_{0\Sigma}F^{\Sigma\mu\nu} - \frac{1}{16}\epsilon^{\mu\nu\rho\sigma\lambda\tau}(\eta_{\Lambda\Sigma}F_{\rho\sigma}^\Lambda F_{\lambda\tau}^\Sigma - 2mB_{\rho\sigma}F_{\lambda\tau}^0 + m^2B_{\rho\sigma}B_{\lambda\tau}), \quad (\text{A.1})$$

$$0 = D_\nu\left(e^{-2\sigma}\mathcal{N}_{\Lambda\Sigma}\widehat{F}^{\Sigma\nu\mu}\right) + 2P^{\mu I\alpha}(L^{-1})_{I\Sigma}f_\Lambda^\Sigma{}_\Pi L^\Pi{}_\alpha - \frac{1}{8}\epsilon^{\mu\rho\sigma\nu\lambda\tau}H_{\rho\sigma\nu}(mB_{\lambda\tau}\delta_{\Lambda 0} + \eta_{\Lambda\Sigma}F_{\lambda\tau}^\Sigma), \quad (\text{A.2})$$

$$0 = R_{\mu\nu} - 4\partial_\mu\sigma\partial_\nu\sigma - P_\mu^{I\alpha}P_{\nu I\alpha} - g_{\mu\nu}V - e^{-2\sigma}\mathcal{N}_{\Lambda\Sigma}\left(\widehat{F}_{\mu\rho}^\Lambda\widehat{F}_{\nu}^{\Sigma\rho} - \frac{1}{8}g_{\mu\nu}\widehat{F}_{\rho\lambda}^\Lambda\widehat{F}^{\Sigma\rho\lambda}\right) - \frac{9}{16}e^{4\sigma}\left(H_{\mu\rho\lambda}H_{\nu}^{\rho\lambda} - \frac{1}{6}g_{\mu\nu}H_{\rho\lambda\tau}H^{\rho\lambda\tau}\right), \quad (\text{A.3})$$

$$\begin{aligned}
0 &= D_\mu D^\mu \sigma - \frac{e^{2\sigma}}{2} \left[\frac{1}{9} A^2 + B_i B^i + C_{Ii} C^{Ii} + 4 D_{Ii} D^{Ii} \right] - 6m e^{-6\sigma} \mathcal{N}_{00} \\
&\quad + 4m e^{-2\sigma} \left(\frac{A}{3} L_{00} - L_{0i} B^i \right) - \frac{e^{-2\sigma}}{4} \mathcal{N}_{\Lambda\Sigma} \hat{F}_{\mu\nu}^\Lambda \hat{F}^{\Sigma\mu\nu} + \frac{3e^{4\sigma}}{16} H_{\mu\nu\rho} H^{\mu\nu\rho}, \quad (\text{A.4})
\end{aligned}$$

$$\begin{aligned}
0 &= \frac{1}{2} D_\mu P^\mu_{I0} + e^{2\sigma} (B_i C_{iI} + K_{iIJ} D^{Ji}) - 2m e^{-2\sigma} L_{0i} C_{Ii} \\
&\quad - e^{-2\sigma} L_{0\Lambda} L_{I\Sigma} \hat{F}_{\mu\nu}^\Lambda \hat{F}^{\Sigma\mu\nu}, \quad (\text{A.5})
\end{aligned}$$

$$\begin{aligned}
0 &= \frac{1}{2} D_\mu P^\mu_{Ii} + 2m e^{-2\sigma} (L_{00} C_{Ii} - 2\varepsilon_{ijk} L_{0j} D_{Ik}) - e^{-2\sigma} L_{i\Lambda} L_{I\Sigma} \hat{F}_{\mu\nu}^\Lambda \hat{F}^{\Sigma\mu\nu} \\
&\quad + e^{2\sigma} \left[\frac{A}{3} C_{Ii} + \varepsilon_{ijk} (B_j D_{Ik} + C_{Jj} K_{kIJ}) \right] \quad (\text{A.6})
\end{aligned}$$

with

$$K_{iIJ} = f_\Lambda{}^\Gamma{}_\Sigma L^\Lambda{}_i (L^{-1})_{I\Gamma} L^\Sigma{}_J. \quad (\text{A.7})$$

B Derivation of $SO(2) \times SO(2)$ symmetric solution

In this appendix, we provide a detailed derivation of $SO(2) \times SO(2)$ symmetric $AdS_4 \times \Sigma$ solution studied in section 3.

B.1 Bosonic ansatz

For convenience, we also repeat the ansatz for the six-dimensional metric here

$$ds_6^2 = f(r) ds_{AdS_4}^2 + h_1(r) dr^2 + h_2(r) d\theta^2 \quad (\text{B.1})$$

where the metric on AdS_4 with unit radius is given by

$$ds_{AdS_4}^2 = \frac{1}{\rho^2} (dx_{1,2}^2 + d\rho^2) \quad (\text{B.2})$$

with $dx_{1,2}^2 = \eta_{mn} dx^m dx^n$, $m, n = 0, 1, 2$ being the flat metric on the three-dimensional Minkowski space Mkw_3 . r and θ are respectively radial and angular coordinates on the topological disk Σ . The six-dimensional curved and flat space-time indices will be split into $\mu = (m, \rho, r, \theta)$ and $\hat{\mu} = (\hat{m}, \hat{\rho}, \hat{r}, \hat{\theta})$, respectively.

With the following vielbein one-forms

$$\begin{aligned}
e_{(1)}^{\hat{m}} &= \frac{\sqrt{f(r)}}{\rho} dx^m, & e_{(1)}^{\hat{\rho}} &= \frac{\sqrt{f(r)}}{\rho} d\rho, \\
e_{(1)}^{\hat{r}} &= \sqrt{h_1(r)} dr, & e_{(1)}^{\hat{\theta}} &= \sqrt{h_2(r)} d\theta,
\end{aligned} \quad (\text{B.3})$$

we can straightforwardly compute all non-vanishing components of the spin connections

$$\begin{aligned}\omega_{(1)}^{\hat{m}\hat{\rho}} &= -\frac{1}{\sqrt{f}}e^{\hat{m}}, & \omega_{(1)}^{\hat{m}\hat{r}} &= \frac{f'}{2f\sqrt{h_1}}e^{\hat{m}}, \\ \omega_{(1)}^{\hat{\rho}\hat{r}} &= \frac{f'}{2f\sqrt{h_1}}e^{\hat{\rho}}, & \omega_{(1)}^{\hat{\theta}\hat{r}} &= \frac{h'_2}{2h_2\sqrt{h_1}}e^{\hat{\theta}}.\end{aligned}\quad (\text{B.4})$$

From now on, we will use primes to denote r -derivatives and mostly suppress arguments of the r -dependent functions for convenience.

For supersymmetric solutions with $SO(2) \times SO(2)$ symmetry, non-vanishing components of the gauge fields are chosen to be

$$A_\theta^3 = A_1(r) \quad \text{and} \quad A_\theta^6 = A_2(r). \quad (\text{B.5})$$

The corresponding two-form field strengths are then given by

$$F_{r\theta}^3 = \frac{A'_1}{2} \quad \text{and} \quad F_{r\theta}^6 = \frac{A'_2}{2}. \quad (\text{B.6})$$

The non-vanishing scalar fields are given by the dilaton $\sigma = \sigma(r)$ and the two $SO(2) \times SO(2)$ singlet scalars $\phi_1 = \phi_1(r)$ and $\phi_2 = \phi_2(r)$ from $SO(4, 3)/(SO(4) \times SO(3))$ given in (3.1).

Setting the two-form field $B_{\mu\nu}$ to zero, we find that the field equation of the two-form field (A.1) reduces to

$$0 = me^{-2\sigma} \mathcal{N}_{0\Sigma} F^{\Sigma\mu\nu} \quad (\text{B.7})$$

which gives

$$\begin{aligned}0 &= \mathcal{N}_{03} F^{3r\theta} + \mathcal{N}_{06} F^{6r\theta} \\ &= A'_1 \sinh \phi_1 \sinh 2\phi_2 - A'_2 \cosh^2 \phi_2 \sinh 2\phi_1.\end{aligned}\quad (\text{B.8})$$

This implies $\phi_1 = 0$ for the two-form field to be consistently truncated out.

With $\phi_1 = 0$ and $H_{\mu\nu\rho} = 0$ as $B_{\mu\nu} = 0$, the gauge field equation (A.2) leads to

$$A''_1 = 2\phi'_2 A'_2 - \left(\frac{4f'}{f} - \frac{h'_1}{h_1} - \frac{h'_2}{h_2} - 4\sigma' \right) \frac{A'_1}{2}, \quad (\text{B.9})$$

$$A''_2 = 2\phi'_2 A'_1 - \left(\frac{4f'}{f} - \frac{h'_1}{h_1} - \frac{h'_2}{h_2} - 4\sigma' \right) \frac{A'_2}{2}. \quad (\text{B.10})$$

The most general solution to these equations is given by

$$A'_1 = (a_1 e^{4\phi_2} + a_2) e^{2\sigma - 2\phi_2} \frac{\sqrt{h_1 h_2}}{2f^2}, \quad (\text{B.11})$$

$$\text{and} \quad A'_2 = (a_1 e^{4\phi_2} - a_2) e^{2\sigma - 2\phi_2} \frac{\sqrt{h_1 h_2}}{2f^2} \quad (\text{B.12})$$

for two integration constants a_1 and a_2 .

B.2 BPS conditions

In order to find supersymmetric solutions, we consider first-order BPS equations derived from the supersymmetry transformations of fermionic fields. For later convenience, we first introduce some notations for the fermion-shift matrices as follows

$$S_{[AB]} = \frac{1}{24} [Ae^\sigma + 6me^{-3\sigma}(L^{-1})_{00}] \epsilon_{AB}, \quad (\text{B.13})$$

$$S_{(AB)} = \frac{1}{8} [B_i e^\sigma - 2me^{-3\sigma}(L^{-1})_{i0}] \sigma_{AB}^i, \quad (\text{B.14})$$

$$N_{[AB]} = \frac{1}{24} [Ae^{-\sigma} - 18me^{-3\sigma}(L^{-1})_{00}] \epsilon_{AB}, \quad (\text{B.15})$$

$$N_{(AB)} = \frac{1}{8} [B_i e^\sigma + 6me^{-3\sigma}(L^{-1})_{i0}] \sigma_{AB}^i. \quad (\text{B.16})$$

We begin with the gravitino variation $\delta\psi_{\mu A}$ along m , ρ , r , and θ directions. These are respectively given by

$$\begin{aligned} 0 = & \partial_m \epsilon_A - \frac{1}{2\sqrt{f}} \Gamma_m \Gamma_{\hat{\rho}} \epsilon_A + \frac{f'}{4f\sqrt{h_1}} \Gamma_m \Gamma_{\hat{r}} \epsilon_A - (S_{[AB]} + S_{(AB)} \Gamma_7) \Gamma_m \epsilon^B \\ & + \frac{i}{8} e^{-\sigma} (T_{[AB]} \Gamma_7 + T_{(AB)}) \Gamma_m \Gamma^{\hat{r}\hat{\theta}} \epsilon^B, \end{aligned} \quad (\text{B.17})$$

$$\begin{aligned} 0 = & \partial_\rho \epsilon_A + \frac{f'}{4f\sqrt{h_1}} \Gamma_\rho \Gamma_{\hat{r}} \epsilon_A - (S_{[AB]} + S_{(AB)} \Gamma_7) \Gamma_\rho \epsilon^B \\ & + \frac{i}{8} e^{-\sigma} (T_{[AB]} \Gamma_7 + T_{(AB)}) \Gamma_\rho \Gamma^{\hat{r}\hat{\theta}} \epsilon^B, \end{aligned} \quad (\text{B.18})$$

$$\begin{aligned} 0 = & \partial_r \epsilon_A + Q_{rAB} \epsilon^B - (S_{[AB]} + S_{(AB)} \Gamma_7) \Gamma_r \epsilon^B \\ & - \frac{3i}{8} \sqrt{h_1} e^{-\sigma} (T_{[AB]} \Gamma_7 + T_{(AB)}) \Gamma^{\hat{\theta}} \epsilon^B, \end{aligned} \quad (\text{B.19})$$

$$\begin{aligned} 0 = & \partial_\theta \epsilon_A + Q_{\theta AB} \epsilon^B + \frac{h'_2}{4h_2\sqrt{h_1}} \Gamma_\theta \Gamma_{\hat{r}} \epsilon_A - (S_{[AB]} + S_{(AB)} \Gamma_7) \Gamma_\theta \epsilon^B \\ & + \frac{3i}{8} \sqrt{h_2} e^{-\sigma} (T_{[AB]} \Gamma_7 + T_{(AB)}) \Gamma^{\hat{r}} \epsilon^B. \end{aligned} \quad (\text{B.20})$$

In these equations, we have also introduced the notations $T_{[AB]}$, $T_{(AB)}$, and T^I for non-vanishing components of the dressed field strength tensors via

$$T_{[AB]} = \epsilon_{AB} (L^{-1})_{0\Lambda} F_{\hat{r}\hat{\theta}}^\Lambda, \quad (\text{B.21})$$

$$T_{(AB)} = \sigma_{AB}^i (L^{-1})_{i\Lambda} F_{\hat{r}\hat{\theta}}^\Lambda, \quad (\text{B.22})$$

$$T^I = (L^{-1})^I_{\Lambda} F_{\hat{r}\hat{\theta}}^\Lambda. \quad (\text{B.23})$$

The next condition from $\delta\chi_A$ yields

$$0 = \frac{\sigma'}{2} \Gamma^r \epsilon_A + (N_{[AB]} - N_{(AB)} \Gamma_7) \epsilon^B + \frac{i}{8} e^{-\sigma} (T_{[AB]} \Gamma_7 - T_{(AB)}) \Gamma^{\hat{r}\hat{\theta}} \epsilon^B \quad (\text{B.24})$$

while the supersymmetry transformation of the gauginos $\delta\lambda_A^I$ gives rise to

$$\begin{aligned} 0 = & P_r^I{}_i \Gamma^r \sigma_{AB}^i \epsilon^B - P_r^I{}_0 \Gamma_7 \Gamma^r \epsilon_A - e^\sigma (2i\Gamma_7 D^I{}_i + C^I{}_i) \sigma_{AB}^i \epsilon^B \\ & - 2me^{-3\sigma} (L^{-1})^I{}_0 \Gamma_7 \epsilon_A - ie^{-\sigma} T^I \Gamma^{\hat{r}\hat{\theta}} \epsilon_A. \end{aligned} \quad (\text{B.25})$$

Recall that the Killing spinors take the form of

$$\epsilon_A = n_A \vartheta \otimes \eta \quad (\text{B.26})$$

with n_A being a constant object in the spinor representation of $SO(3)_R$. η is a two-component spinor depending on r and θ coordinates while ϑ is a four-component Killing spinor on AdS_4 satisfying

$$\nabla_{\hat{a}}^{AdS_4} \vartheta = \frac{is}{2} \gamma_* \gamma_{\hat{a}} \vartheta \quad (\text{B.27})$$

for $\hat{a} = (\hat{m}, \hat{\rho}) = 0, 1, 2, 3$. The real parameter $s = \pm 1$ is an arbitrary sign choice. $\gamma_{\hat{a}}$ are four-dimensional 4×4 gamma matrices satisfying the Clifford algebra

$$\{\gamma_{\hat{a}}, \gamma_{\hat{b}}\} = 2\eta_{\hat{a}\hat{b}} \mathbb{1}_4 \quad \text{for} \quad \eta_{\hat{a}\hat{b}} = \text{diag}(-1, 1, 1, 1). \quad (\text{B.28})$$

The four-dimensional chirality matrix is defined by $\gamma_* = i\gamma^0\gamma^1\gamma^2\gamma^3$ with $\gamma_*^2 = \mathbb{1}_4$. We can also decompose the six-dimensional gamma matrices in terms of $\gamma_{\hat{a}}$ and γ_* as

$$\Gamma_{\hat{a}} = i\gamma_* \gamma_{\hat{a}} \otimes \sigma_3, \quad \Gamma_{\hat{r}} = \mathbb{1}_4 \otimes \sigma_1, \quad \Gamma_{\hat{\theta}} = \mathbb{1}_4 \otimes \sigma_2 \quad (\text{B.29})$$

in which σ_1 , σ_2 , and σ_3 are Pauli matrices. In this basis, the six-dimensional chirality matrix then takes the form

$$\Gamma_7 = i\gamma_* \otimes \sigma_3. \quad (\text{B.30})$$

With the supersymmetry parameter (B.26) and six-dimensional gamma matrices in (B.29), the first two BPS equations given in (B.17) and (B.18) reduce to a single equation

$$\begin{aligned} 0 = & \frac{n_A}{\sqrt{f}} (\nabla_{\hat{a}}^{AdS_4} \vartheta) \otimes \eta + \frac{f'}{4f\sqrt{h_1}} n_A (i\gamma_* \gamma_{\hat{a}} \vartheta) \otimes (i\sigma_2 \eta) \\ & - S_{[AB]} n^B (i\gamma_* \gamma_{\hat{a}} \vartheta) \otimes (\sigma_3 \eta) + S_{(AB)} n^B (\gamma_{\hat{a}} \vartheta) \otimes \eta \\ & + \frac{e^{-\sigma}}{8} T_{[AB]} n^B (\gamma_{\hat{a}} \vartheta) \otimes (\sigma_3 \eta) - \frac{e^{-\sigma}}{8} T_{(AB)} n^B (i\gamma_* \gamma_{\hat{a}} \vartheta) \otimes \eta \end{aligned} \quad (\text{B.31})$$

in which we have expressed the supersymmetry parameters in terms of two- and four-component spinors. With the Killing spinor equation (B.27), we can split this

equation into two parts with and without the four-dimensional chirality matrix acting on ϑ

$$0 = i\gamma_*\gamma_{\hat{a}}\vartheta \otimes \left\{ \frac{s}{2\sqrt{f}}n_A\eta + \frac{f'}{4f\sqrt{h_1}}n_A(i\sigma_2)\eta - S_{[AB]}n^B\sigma_3\eta - \frac{e^{-\sigma}}{8}T_{(AB)}n^B\eta \right\} \\ + \gamma_{\hat{a}}\vartheta \otimes \left\{ S_{(AB)}n^B\eta + \frac{e^{-\sigma}}{8}T_{[AB]}n^B\sigma_3\eta \right\}. \quad (\text{B.32})$$

These two parts are linearly independent and have to vanish separately.

Repeating the same procedure for the remaining conditions, we find the following set of BPS equations on the two-component spinor η

$$0 = \frac{s}{2\sqrt{f}}n_A\eta + \frac{f'}{4f\sqrt{h_1}}n_A(i\sigma_2)\eta - S_{[AB]}n^B\sigma_3\eta - \frac{e^{-\sigma}}{8}T_{(AB)}n^B\eta, \quad (\text{B.33})$$

$$0 = \frac{1}{\sqrt{h_1}}(n_A\partial_r\eta + Q_{rAB}n^B\eta) - S_{[AB]}n^B\sigma_1\eta - \frac{3}{8}e^{-\sigma}T_{(AB)}n^B(i\sigma_2)\eta, \quad (\text{B.34})$$

$$0 = \frac{i}{\sqrt{h_2}}(n_A\partial_\theta\eta + Q_{\theta AB}n^B\eta) + \frac{h'_2}{4h_2\sqrt{h_1}}n_A\sigma_3\eta - S_{[AB]}n^B(i\sigma_2)\eta \\ - \frac{3}{8}e^{-\sigma}T_{(AB)}n^B(\sigma_1)\eta, \quad (\text{B.35})$$

$$0 = \frac{\sigma'}{2\sqrt{h_1}}n_A\sigma_1\eta + N_{[AB]}n^B\eta + \frac{e^{-\sigma}}{8}T_{(AB)}n^B\sigma_3\eta, \quad (\text{B.36})$$

$$0 = \frac{1}{\sqrt{h_1}}P_r^I{}_i\sigma_{AB}^in^B\sigma_1\eta - e^\sigma C^I{}_i\sigma_{AB}^in^B\eta + e^{-\sigma}T^In_A\sigma_3\eta, \quad (\text{B.37})$$

$$0 = \frac{1}{\sqrt{h_1}}P_r^I{}_0n_A(i\sigma_2)\eta + 2ie^\sigma D^I{}_i\sigma_{AB}^in^B\sigma_3\eta + 2me^{-3\sigma}(L^{-1})^I{}_0n_A\sigma_3\eta, \quad (\text{B.38})$$

$$0 = S_{(AB)}n^B\eta + \frac{e^{-\sigma}}{8}T_{[AB]}n^B\sigma_3\eta, \quad (\text{B.39})$$

$$0 = S_{(AB)}n^B\eta - \frac{3}{8}e^{-\sigma}T_{[AB]}n^B\sigma_3\eta, \quad (\text{B.40})$$

$$0 = N_{(AB)}n^B\eta + \frac{e^{-\sigma}}{8}T_{[AB]}n^B\sigma_3\eta. \quad (\text{B.41})$$

It can be readily verified that the difference between (B.39) and (B.40) leads to an algebraic constraint

$$T_{[AB]}n^B\sigma_3\eta = 0. \quad (\text{B.42})$$

Substituting this constraint into the last three equations, we find that

$$S_{(AB)}n^B\eta = 0 \quad \text{and} \quad N_{(AB)}n^B\eta = 0. \quad (\text{B.43})$$

Since $n^B\eta \neq 0$, we need to impose the following conditions

$$T_{[AB]} = S_{(AB)} = N_{(AB)} = 0 \quad (\text{B.44})$$

in order to obtain supersymmetric $AdS_4 \times \Sigma$ solutions.

With the coset representative given in (3.1) together with the two-form field strengths (B.6), the explicit form of these quantities reads

$$S_{(AB)} = \frac{m}{4} e^{-3\sigma} \sinh \phi_1 \sinh \phi_2 \sigma_{AB}^3, \quad (B.45)$$

$$N_{(AB)} = -\frac{3m}{4} e^{-3\sigma} \sinh \phi_1 \sinh \phi_2 \sigma_{AB}^3, \quad (B.46)$$

$$T_{[AB]} = -A'_2 \sinh \phi_1 \epsilon_{AB}. \quad (B.47)$$

We see again that ϕ_1 must vanish in consistent with the previous result obtained from truncating out the two-form field. It should also be noted that with $\phi_1 = 0$, the quantities $P_r^I{}_0$, $D^I{}_i$, and $(L^{-1})^I{}_0$ also vanish, so the BPS equation in (B.38) is trivially satisfied.

The composite connections $Q_{rAB} = 0$ and $Q_{\theta AB}$ take the form of

$$Q_{\theta AB} = -\frac{i}{2} g_1 A_\theta^3 \sigma_{AB}^3 = -\frac{i}{2} g_1 A_1(r) \sigma_{AB}^3. \quad (B.48)$$

We also note that these involve only the $SO(2)_R \subset SO(3)_R$ gauge field A^3 since the supersymmetry parameters are charged exclusively under $SO(3)_R$. Following [52], we assume a definite charge of the two-component spinor $\eta(r, \theta)$ under the $U(1)_\theta$ isometry. This leads to the ansatz for η of the form

$$\eta(r, \theta) = e^{iq\theta} \hat{\eta}(r) \quad (B.49)$$

with a constant q . By imposing the following projector

$$\sigma_{AB}^3 n^B = -n_A, \quad (B.50)$$

we find that the combination $(n_A \partial_\theta \eta + Q_{\theta AB} n^B \eta) = (in_A q \eta - \frac{i}{2} g_1 A_1 \sigma_{AB}^3 n^B \eta)$ in (B.35) is invariant under the transformations

$$A_1 \rightarrow A_1 - \frac{2\alpha_0}{g_1} \quad \text{and} \quad \eta \rightarrow e^{i\alpha_0 \theta} \eta \quad (B.51)$$

where α_0 is an arbitrary constant. It is then convenient to define

$$g_1 \hat{A}_1 = 2q + g_1 A_1 \quad (B.52)$$

with $\hat{A}'_1 = A'_1$.

With all these and suitable left-multiplications by Pauli matrices, we can subtract (B.33) by (B.36) and combine (B.35) with (B.36) to obtain the following

non-trivial set of the BPS equations

$$0 = \frac{s}{\sqrt{f}} n_A \eta + \frac{1}{\sqrt{h_1}} \left(\frac{f'}{2f} + \sigma' \right) n_A (i\sigma_2) \eta - 2 (S_{[AB]} - N_{[AB]}) n^B \sigma_3 \eta, \quad (\text{B.53})$$

$$0 = \frac{1}{\sqrt{h_1}} n_A \partial_r \eta - S_{[AB]} n^B \sigma_1 \eta - \frac{3}{8} e^{-\sigma} T_{(AB)} n^B (i\sigma_2) \eta, \quad (\text{B.54})$$

$$0 = \frac{g_1 \hat{A}_1}{\sqrt{h_2}} n_A \sigma_1 \eta + \frac{1}{\sqrt{h_1}} \left(\frac{h'_2}{2h_2} + \sigma' \right) n_A (i\sigma_2) \eta - 2 (S_{[AB]} - N_{[AB]}) n^B \sigma_3 \eta \\ + e^{-\sigma} T_{(AB)} n^B \eta, \quad (\text{B.55})$$

$$0 = \frac{\sigma'}{\sqrt{h_1}} n_A \sigma_3 \eta + 2 N_{[AB]} n^B (i\sigma_2) \eta - \frac{1}{4} e^{-\sigma} T_{(AB)} n^B \sigma_1 \eta, \quad (\text{B.56})$$

$$0 = \frac{1}{\sqrt{h_1}} P_r^I \sigma_{AB}^i n^B \sigma_3 \eta - e^\sigma C^I \sigma_{AB}^i n^B (i\sigma_2) \eta - e^{-\sigma} T^I n_A \sigma_1 \eta. \quad (\text{B.57})$$

B.3 The analysis of the BPS equations

We now perform an analysis of all the previously derived conditions and determine the final form of the relevant BPS equations. With $\phi_1 = 0$, the explicit form of non-vanishing components of fermion-shift matrices is given by

$$S_{[AB]} = \frac{1}{4} e^{-3\sigma} (m + g_1 e^{4\sigma} \cosh \phi_2) \epsilon_{AB}, \quad (\text{B.58})$$

$$N_{[AB]} = -\frac{1}{4} e^{-3\sigma} (3m - g_1 e^{4\sigma} \cosh \phi_2) \epsilon_{AB}, \quad (\text{B.59})$$

$$T_{(AB)} = \frac{1}{2\sqrt{h_1 h_2}} (A'_1 \cosh \phi_2 - A'_2 \sinh \phi_2) \sigma_{AB}^3 \quad (\text{B.60})$$

together with

$$P_r^I \sigma_{AB}^i = \phi'_2 \delta_3^I \delta_{i3}, \quad (\text{B.61})$$

$$C^I \sigma_{AB}^i = 2g_1 \sinh \phi_2 \delta_3^I \delta_{i3}, \quad (\text{B.62})$$

$$T^I = \frac{1}{2\sqrt{h_1 h_2}} (A'_2 \cosh \phi_2 - A'_1 \sinh \phi_2) \delta_3^I. \quad (\text{B.63})$$

With these quantities and the projector (B.50), the BPS conditions become

$$0 = \frac{s}{\sqrt{f}} \eta + \frac{1}{\sqrt{h_1}} \left(\frac{f'}{2f} + \sigma' \right) (i\sigma_2) \eta + 2m e^{-3\sigma} \sigma_3 \eta, \quad (\text{B.64})$$

$$0 = \frac{1}{\sqrt{h_1}} \partial_r \eta + \frac{1}{4} e^{-3\sigma} (m + g_1 e^{4\sigma} \cosh \phi_2) \sigma_1 \eta + \frac{3}{8} e^{-\sigma} \mathbb{F}_1 (i\sigma_2) \eta, \quad (\text{B.65})$$

$$0 = \frac{g_1 \hat{A}_1}{\sqrt{h_2}} \sigma_1 \eta + \frac{1}{\sqrt{h_1}} \left(\frac{h'_2}{2h_2} + \sigma' \right) (i\sigma_2) \eta + 2me^{-3\sigma} \sigma_3 \eta - e^{-\sigma} \mathbb{F}_1 \eta, \quad (\text{B.66})$$

$$0 = \frac{\sigma'}{\sqrt{h_1}} \sigma_3 \eta + \frac{1}{2} e^{-3\sigma} (3m - g_1 e^{4\sigma} \cosh \phi_2) (i\sigma_2) \eta + \frac{1}{4} e^{-\sigma} \mathbb{F}_1 \sigma_1 \eta, \quad (\text{B.67})$$

$$0 = \frac{\phi'_2}{\sqrt{h_1}} \sigma_3 \eta - 2g_1 e^{\sigma} \sinh \phi_2 (i\sigma_2) \eta + e^{-\sigma} \mathbb{F}_2 \sigma_1 \eta \quad (\text{B.68})$$

with

$$\mathbb{F}_1 = \frac{1}{2\sqrt{h_1 h_2}} (A'_1 \cosh \phi_2 - A'_2 \sinh \phi_2), \quad (\text{B.69})$$

$$\mathbb{F}_2 = \frac{1}{2\sqrt{h_1 h_2}} (A'_2 \cosh \phi_2 - A'_1 \sinh \phi_2). \quad (\text{B.70})$$

Equation (B.65) determines the explicit form of $\eta(r, \theta)$ while the other four equations are of the form $M^{(x)} \eta = 0$ where the index $x = 1, 2, 3, 4$ respectively labels the BPS equations in (B.64), (B.66), (B.67), and (B.68). The four 2×2 matrices $M^{(x)}$ can be parametrized by

$$M^{(x)} = X_0^{(x)} \mathbb{1}_2 + X_1^{(x)} \sigma_1 + X_2^{(x)} (i\sigma_2) + X_3^{(x)} \sigma_3. \quad (\text{B.71})$$

Following [52], for each matrix $M^{(x)}$, we define the two-component vectors

$$v^{(x)} = \begin{pmatrix} X_1^{(x)} + X_2^{(x)} \\ -X_0^{(x)} - X_3^{(x)} \end{pmatrix} \quad \text{and} \quad w^{(x)} = \begin{pmatrix} X_0^{(x)} - X_3^{(x)} \\ -X_1^{(x)} + X_2^{(x)} \end{pmatrix} \quad (\text{B.72})$$

together with

$$\mathcal{A}^{xy} = \det(v^{(x)} | w^{(y)}), \quad \mathcal{B}^{xy} = \det(v^{(x)} | v^{(y)}), \quad \mathcal{C}^{xy} = \det(w^{(x)} | w^{(y)}). \quad (\text{B.73})$$

The notation $(v|w)$ denotes a 2×2 matrix obtained from a juxtaposition of the two-column vectors v and w . As pointed out in [52], the vanishing of $\mathcal{A}^{xy}$, $\mathcal{B}^{xy}$, and $\mathcal{C}^{xy}$ gives a number of necessary conditions for the existence of a non-trivial solution for η as well as BPS equations for all the other fields. We will solve the conditions from the supergravity and vector multiplets separately by splitting the index x as $(\bar{x}, 4)$ with $\bar{x} = 1, 2, 3$.

Starting from the first matrix $\mathcal{A}^{xy}$ that is asymmetric in xy indices, the vanishing of the diagonal components $\mathcal{A}^{\bar{x}\bar{x}}$ gives the following conditions

$$0 = \frac{1}{f} + \frac{1}{4h_1} \left(\frac{f'}{f} + 2\sigma' \right)^2 - 4m^2 e^{-6\sigma}, \quad (\text{B.74})$$

$$0 = (\mathbb{F}_1)^2 e^{-2\sigma} + \frac{1}{4h_1} \left(\frac{h'_2}{h_2} + 2\sigma' \right)^2 - 4m^2 e^{-6\sigma} - \frac{(g_1 \hat{A}_1)^2}{h_2}, \quad (\text{B.75})$$

$$0 = \frac{16\sigma'^2}{h_1} - 4e^{-6\sigma} (g_1 e^{4\sigma} \cosh \phi_2 - 3m)^2 + (\mathbb{F}_1)^2 e^{-2\sigma}. \quad (\text{B.76})$$

The vanishing of the off-diagonal symmetric components $\mathcal{A}^{\bar{x}\bar{y}} + \mathcal{A}^{\bar{y}\bar{x}}$ for $\bar{x} \neq \bar{y}$ yields

$$0 = \frac{1}{2h_1} \left(\frac{f'}{f} + 2\sigma' \right) \left(\frac{h'_2}{h_2} + 2\sigma' \right) - \frac{2s}{\sqrt{f}} \mathbb{F}_1 e^{-\sigma} - 8m^2 e^{-6\sigma}, \quad (\text{B.77})$$

$$0 = 8m\sigma' + \left(\frac{f'}{f} + 2\sigma' \right) (g_1 e^{4\sigma} \cosh \phi_2 - 3m), \quad (\text{B.78})$$

$$0 = \frac{8m\sigma'}{\sqrt{h_1}} + \frac{1}{\sqrt{h_1}} \left(\frac{h'_2}{h_2} + 2\sigma' \right) (g_1 e^{4\sigma} \cosh \phi_2 - 3m) + \frac{g_1 \hat{A}_1}{\sqrt{h_2}} \mathbb{F}_1 e^{2\sigma} \quad (\text{B.79})$$

while the off-diagonal antisymmetric components $\mathcal{A}^{\bar{x}\bar{y}} - \mathcal{A}^{\bar{y}\bar{x}} = 0$ give

$$0 = \frac{4sm}{\sqrt{f}} e^{-3\sigma} + \frac{g_1 \hat{A}_1}{\sqrt{h_1 h_2}} \left(\frac{f'}{f} + 2\sigma' \right) + 4m \mathbb{F}_1 e^{-4\sigma}, \quad (\text{B.80})$$

$$0 = \frac{8s\sigma'}{\sqrt{f}} + \left(\frac{f'}{f} + 2\sigma' \right) \mathbb{F}_1 e^{-\sigma}, \quad (\text{B.81})$$

$$0 = \frac{4g_1 \hat{A}_1}{\sqrt{h_2}} (g_1 e^{4\sigma} \cosh \phi_2 - 3m) + \frac{1}{\sqrt{h_1}} \left(\frac{h'_2}{h_2} - 6\sigma' \right) \mathbb{F}_1 e^{2\sigma}. \quad (\text{B.82})$$

On the other hand, matrices $\mathcal{B}^{xy}$ and $\mathcal{C}^{xy}$ are both antisymmetric in xy indices. It turns out that the BPS conditions arising from the combinations $\mathcal{B}^{xy} \pm \mathcal{C}^{xy} = 0$ take a simpler form. The combinations $\mathcal{B}^{\bar{x}\bar{y}} + \mathcal{C}^{\bar{x}\bar{y}} = 0$ give the following conditions

$$0 = \frac{e^{-\sigma}}{\sqrt{h_1}} \left(\frac{f'}{f} + 2\sigma' \right) \mathbb{F}_1 + \frac{s}{\sqrt{f h_1}} \left(\frac{h'_2}{h_2} + 2\sigma' \right) + \frac{4m}{\sqrt{h_2}} g_1 \hat{A}_1 e^{-3\sigma}, \quad (\text{B.83})$$

$$0 = m \mathbb{F}_1 e^{-\sigma} - \frac{s}{\sqrt{f}} (g_1 e^{4\sigma} \cosh \phi_2 - 3m), \quad (\text{B.84})$$

$$0 = \frac{2g_1 \hat{A}_1 \sigma'}{\sqrt{h_1 h_2}} - \mathbb{F}_1 (g_1 \cosh \phi_2 - 2m e^{-4\sigma}) \quad (\text{B.85})$$

while the combinations $\mathcal{B}^{\bar{x}\bar{y}} - \mathcal{C}^{\bar{x}\bar{y}} = 0$ lead to

$$0 = \frac{m e^{-3\sigma}}{\sqrt{h_1}} \left(\frac{f'}{f} - \frac{h'_2}{h_2} \right) - \frac{s g_1 \hat{A}_1}{\sqrt{f h_2}}, \quad (\text{B.86})$$

$$0 = \frac{s e^{-\sigma}}{\sqrt{f}} \mathbb{F}_1 - \frac{2\sigma'}{h_1} \left(\frac{f'}{f} + 2\sigma' \right) - 4m e^{-6\sigma} (g_1 e^{4\sigma} \cosh \phi_2 - 3m), \quad (\text{B.87})$$

$$0 = e^{-2\sigma} (\mathbb{F}_1)^2 + \frac{2\sigma'}{h_1} \left(\frac{h'_2}{h_2} + 2\sigma' \right) + 4m e^{-6\sigma} (g_1 e^{4\sigma} \cosh \phi_2 - 3m). \quad (\text{B.88})$$

Accordingly, there are in total fifteen algebraic conditions obtained from the BPS equations of the supergravity multiplet (B.64), (B.66), and (B.67).

Extending this procedure to the BPS equation (B.68) from vector multiplets gives additional BPS conditions derived from the vanishing of $\mathcal{A}^{x4}$, $\mathcal{B}^{\bar{x}4}$, and $\mathcal{C}^{\bar{x}4}$. The first condition obtained from $\mathcal{A}^{44} = 0$ takes the form

$$0 = \frac{(\phi'_2)^2}{h_1} - 4e^{2\sigma} (g_1 \sinh \phi_2)^2 + (\mathbb{F}_2)^2 e^{-2\sigma}. \quad (\text{B.89})$$

The symmetric part $\mathcal{A}^{\bar{x}4} + \mathcal{A}^{4\bar{x}} = 0$ gives

$$0 = 2m\phi'_2 + g_1 e^{4\sigma} \sinh \phi_2 \left(\frac{f'}{f} + 2\sigma' \right), \quad (\text{B.90})$$

$$0 = \frac{2m\phi'_2}{\sqrt{h_1}} + \frac{g_1}{\sqrt{h_1}} e^{4\sigma} \sinh \phi_2 \left(\frac{h'_2}{h_2} + 2\sigma' \right) + \frac{g_1 \hat{A}_1}{\sqrt{h_2}} \mathbb{F}_2 e^{2\sigma}, \quad (\text{B.91})$$

$$0 = \frac{4\sigma'\phi'_2}{h_1} e^{2\sigma} - 4g_1 \sinh \phi_2 (g_1 e^{4\sigma} \cosh \phi_2 - 3m) + \mathbb{F}_1 \mathbb{F}_2 \quad (\text{B.92})$$

while the antisymmetric part $\mathcal{A}^{\bar{x}4} - \mathcal{A}^{4\bar{x}} = 0$ results in

$$0 = \frac{2s\phi'_2}{\sqrt{f}} + \left(\frac{f'}{f} + 2\sigma' \right) \mathbb{F}_2 e^{-\sigma}, \quad (\text{B.93})$$

$$0 = \frac{4g_1^2 \hat{A}_1}{\sqrt{h_2}} e^{2\sigma} \sinh \phi_2 + \frac{1}{\sqrt{h_1}} \left(\frac{h'_2}{h_2} + 2\sigma' \right) \mathbb{F}_2 - \frac{2\phi'_2}{\sqrt{h_1}} \mathbb{F}_1, \quad (\text{B.94})$$

$$0 = g_1 \mathbb{F}_1 e^{4\sigma} \sinh \phi_2 - (g_1 e^{4\sigma} \cosh \phi_2 - 3m) \mathbb{F}_2. \quad (\text{B.95})$$

Moreover, the combinations $\mathcal{B}^{\bar{x}4} + \mathcal{C}^{\bar{x}4} = 0$ and $\mathcal{B}^{\bar{x}4} - \mathcal{C}^{\bar{x}4} = 0$ respectively lead to the following sets of extra conditions

$$0 = \frac{sg_1}{\sqrt{f}} e^{5\sigma} \sinh \phi_2 - m \mathbb{F}_2, \quad (\text{B.96})$$

$$0 = 2g_1 \mathbb{F}_1 \sinh \phi_2 + 2m \mathbb{F}_2 e^{-4\sigma} - \frac{g_1 \hat{A}_1 \phi'_2}{\sqrt{h_1 h_2}}, \quad (\text{B.97})$$

$$0 = 4\mathbb{F}_2 \sigma' - \mathbb{F}_1 \phi'_2 \quad (\text{B.98})$$

and

$$0 = \frac{\phi'_2}{h_1} \left(\frac{f'}{f} + 2\sigma' \right) + 8mg_1 e^{-2\sigma} \sinh \phi_2 - \frac{2se^{-\sigma}}{\sqrt{f}} \mathbb{F}_2, \quad (\text{B.99})$$

$$0 = \frac{\phi'_2}{h_1} \left(\frac{h'_2}{h_2} + 2\sigma' \right) + 8mg_1 e^{-2\sigma} \sinh \phi_2 + 2e^{-2\sigma} \mathbb{F}_1 \mathbb{F}_2, \quad (\text{B.100})$$

$$0 = \phi'_2 (g_1 e^{4\sigma} \cosh \phi_2 - 3m) - 4g_1 e^{4\sigma} \sigma' \sinh \phi_2. \quad (\text{B.101})$$

We now solve all these conditions in order to find supersymmetric $AdS_4 \times \Sigma$ solutions. With $\mathbb{F}_1$ and $\mathbb{F}_2$ given in (B.69) and (B.70) together with the explicit

forms of A'_1 and A'_2 given in (B.11) and (B.12), we can solve equation (B.95) and find the solution for ϕ_2 of the form

$$\phi_2 = \ln \left[\frac{(a_1 - a_2)g_1 e^{4\sigma} + \kappa \sqrt{(a_1 - a_2)^2 g_1^2 e^{8\sigma} + 36a_1 a_2 m^2}}{6a_1 m} \right] \quad (\text{B.102})$$

where $\kappa = \pm 1$. Similarly, equations (B.84) and (B.76) can be solved for f and h_1 as functions of σ and ϕ_2 as follows

$$f = \frac{m^{\frac{2}{3}} e^{\frac{2}{3}(\sigma - \phi_2)} (a_1 e^{2\phi_2} + a_2)^{\frac{2}{3}}}{(4s (g_1 e^{4\sigma} \cosh \phi_2 - 3m))^{\frac{2}{3}}}, \quad (\text{B.103})$$

$$h_1 = \frac{16m^{\frac{8}{3}} (a_1 e^{2\phi_2} + a_2)^{\frac{2}{3}} (g_1 e^{4\sigma} \cosh \phi_2 - 3m)^{-2} e^{6\sigma} (\sigma')^2}{4m^{\frac{8}{3}} (a_1 e^{2\phi_2} + a_2)^{\frac{2}{3}} - e^{\frac{2}{3}(8\sigma + \phi_2)} (4s (g_1 e^{4\sigma} \cosh \phi_2 - 3m))^{\frac{2}{3}}}. \quad (\text{B.104})$$

We now determine the function $\hat{A}_1$ defined in (B.52). From (B.85), we find the following expression for $\sqrt{h_1 h_2}$

$$\sqrt{h_1 h_2} = \frac{8g_1 f^2 \hat{A}_1 e^{2\sigma + \phi_2} \sigma'}{(a_1 e^{2\phi_2} + a_2) (g_1 e^{4\sigma} \cosh \phi_2 - 2m)}. \quad (\text{B.105})$$

With $A'_1 = \hat{A}'_1$, we also find another expression for $\sqrt{h_1 h_2}$ from (B.11)

$$\sqrt{h_1 h_2} = \frac{2f^2 \hat{A}'_1 e^{2\phi_2 - 2\sigma}}{a_1 e^{4\phi_2} + a_2}. \quad (\text{B.106})$$

These two equations together with all the previous results lead to an ordinary differential equation for $\hat{A}_1$ of the form

$$\hat{A}'_1 = \frac{72e^{4\sigma} a_1 g_1 m^2 (a_1 \mathbb{K}^4 + a_2) [(a_1 - a_2)g_1 e^{4\sigma} \mathbb{K} + 6a_2 m]^{-1} \sigma' \hat{A}_1}{(a_1 - a_2)g_1^2 e^{8\sigma} \mathbb{K} + 3(a_1 + a_2)g_1 m e^{4\sigma} - 12a_1 m^2 \mathbb{K}} \quad (\text{B.107})$$

where

$$\mathbb{K} = e^{\phi_2} = \frac{(a_1 - a_2)g_1 e^{4\sigma} + \kappa \sqrt{(a_1 - a_2)^2 g_1^2 e^{8\sigma} + 36a_1 a_2 m^2}}{6a_1 m}. \quad (\text{B.108})$$

Solving this equation leads to the following solution

$$\begin{aligned} \hat{A}_1 &= \frac{\mathbb{C}}{6} e^{8\sigma} \sqrt{\frac{4k^4 - (5a_1 + a_2)(a_1 + 5a_2)g_1^2 k^2 + g_1^4 \Delta a^4}{a_1 a_2}} \\ &\times \left\{ \frac{[u^2 (k^2 - g_1^2 \Delta a^2) + 8a_1 a_2 (g_1^2 \Delta a^2 - 4k^2)] / (6a_1 a_2 y)}{2\sqrt{2}g_1 k u \Delta a^2 + g_1^2 \Delta a^2 (u^2 - 8a_1 a_2) + k^2 (u^2 - 32a_1 a_2)} \right\}^{\frac{\kappa(y+a_1+a_2)}{2\sqrt{2}u}} \\ &\times \left\{ \frac{[v^2 (k^2 - g_1^2 \Delta a^2) + 8a_1 a_2 (g_1^2 \Delta a^2 - 4k^2)] / (48a_1 a_2)}{2\sqrt{2}g_1 k v \Delta a^2 - g_1^2 \Delta a^2 (v^2 - 8a_1 a_2) - k^2 (v^2 - 32a_1 a_2)} \right\}^{\frac{\kappa(y-a_1-a_2)}{2\sqrt{2}v}} \\ &\times \left\{ \frac{e^{8\sigma} g_1 v \Delta a^2 (y - a_1 - a_2) (k^2 - \Delta a^2 g_1^2)}{48\sqrt{2}a_1 a_2} \right\}^{\frac{\kappa(y-a_1-a_2)}{2\sqrt{2}v}} \end{aligned} \quad (\text{B.109})$$

with an integration constant $\mathbb{C}$. In this equation, we have defined the following quantities

$$\Delta a^2 = (a_1 - a_2)^2 \quad (\text{B.110})$$

and

$$u = \sqrt{a_1^2 + 18a_1a_2 + a_2^2 + (c_1 + c_2)y}, \quad (\text{B.111})$$

$$v = \sqrt{a_1^2 + 18a_1a_2 + a_2^2 - (c_1 + c_2)y}, \quad (\text{B.112})$$

$$k = \sqrt{(a_1 - a_2)^2 g_1^2 e^{8\sigma} + 36a_1a_2 m^2}, \quad (\text{B.113})$$

$$y = \sqrt{a_1^2 + 34a_1a_2 + a_2^2}. \quad (\text{B.114})$$

Substituting these solutions into (B.105), we can also find the solution for h_2 in terms of the other ones. However, the explicit form of this solution is far more complicated, so we refrain from giving it here. Moreover, finding the solution for A_2 is much more difficult since all the previously obtained results lead to a highly complicated differential equation for A'_2 . The situation is similar to that of [57] in which the $AdS_5 \times \Sigma$ solution with $SO(2) \times SO(2)$ symmetry has been studied in $N = 2$ seven-dimensional gauged supergravity. In any case, it should also be noted that A_2 does not appear in any BPS equations in which only A'_1 and A'_2 appear. In particular, we have verified that all the BPS conditions are satisfied by the above solution without the explicit solution for A_2 .

B.4 A simplified solution

To make subsequent analysis of the solutions more traceable, we will further simplify the solution by setting

$$a_1 = -a_2. \quad (\text{B.115})$$

It should be noted that a more obvious choice with $a_1 = a_2$ leads to a trivial solution for equation (B.95) namely $\phi_2 = 0$. With $\phi_2 = 0$ and $a_1 = a_2$, we find that $A'_2 = 0$, see (B.12), giving rise to constant A_2 that can be set to zero. Therefore, all the fields from vector multiplets are truncated out recovering the BPS equations studied in [54] for pure $F(4)$ gauged supergravity up to notational and conventional differences.

In the case of $a_1 = -a_2 = b$, we find non-vanishing components of the dressed field strengths given by

$$\mathbb{F}_1 = \frac{be^{2\sigma}}{2f^2} \sinh \phi_2 \quad \text{and} \quad \mathbb{F}_2 = \frac{be^{2\sigma}}{2f^2} \cosh \phi_2. \quad (\text{B.116})$$

The previously obtained solution now takes a much simpler form

$$\phi_2 = \ln \left[\frac{g_1 e^{4\sigma} + \kappa \sqrt{g_1^2 e^{8\sigma} - 9m^2}}{3m} \right], \quad (\text{B.117})$$

$$f = \frac{b^{\frac{2}{3}} m^{\frac{2}{3}} e^{\frac{2\sigma}{3}}}{s^{\frac{2}{3}} 2^{\frac{2}{3}} (g_1^2 e^{8\sigma} - 9m^2)^{\frac{1}{3}}}, \quad (\text{B.118})$$

$$h_1 = \frac{9 \times 2^{\frac{10}{3}} m^2 e^{6\sigma} \left(\frac{m^4 b}{s} \right)^{\frac{2}{3}} (\sigma')^2}{(g_1^2 e^{8\sigma} - 9m^2)^2 \left[2^{\frac{4}{3}} \left(\frac{m^4 b}{s} \right)^{\frac{2}{3}} - e^{\frac{16\sigma}{3}} (g_1^2 e^{8\sigma} - 9m^2)^{\frac{1}{3}} \right]}, \quad (\text{B.119})$$

$$h_2 = \frac{9C^2 g_1^2 m^2 e^{\frac{2\sigma}{3}} \left[2^{\frac{4}{3}} \left(\frac{m^4 b}{s} \right)^{\frac{2}{3}} - e^{\frac{16\sigma}{3}} (g_1^2 e^{8\sigma} - 9m^2)^{\frac{1}{3}} \right]}{4 (g_1^2 e^{8\sigma} - 9m^2)^{\frac{1}{3}}}, \quad (\text{B.120})$$

$$\hat{A}_1 = C (g_1^2 e^{8\sigma} - 6m^2) \quad (\text{B.121})$$

where C is an integration constant. It can also be verified that all of the BPS equations and the field equations are satisfied for

$$g_1^2 e^{8\sigma} - 9m^2 > 0. \quad (\text{B.122})$$

In addition, equation (B.12) leads to the following differential equation

$$A_2' = \frac{4g_1 C e^{4\sigma} (2g_1^2 e^{8\sigma} - 9m^2) \sigma'}{\sqrt{g_1^2 e^{8\sigma} - 9m^2}} \quad (\text{B.123})$$

which can be readily solved by

$$A_2 = g_1 C e^{4\sigma} \sqrt{g_1^2 e^{8\sigma} - 9m^2} + c_2 \quad (\text{B.124})$$

for another integration constant c_2 . We also note an explicit form of A_1 which reads

$$A_1 = C (g_1^2 e^{8\sigma} - 6m^2) - \frac{2q}{g_1}. \quad (\text{B.125})$$

Finally, the BPS equations (B.64), (B.66), (B.67), and (B.68) are satisfied by the following form of the two-component spinor

$$\eta = e^{iq\theta} Y \begin{pmatrix} \sqrt{2^{\frac{2}{3}} \left(\frac{m^4 b}{s} \right)^{\frac{1}{3}} - s e^{\frac{8\sigma}{3}} (g_1^2 e^{8\sigma} - 9m^2)^{\frac{1}{6}}} \\ \sqrt{2^{\frac{2}{3}} \left(\frac{m^4 b}{s} \right)^{\frac{1}{3}} + s e^{\frac{8\sigma}{3}} (g_1^2 e^{8\sigma} - 9m^2)^{\frac{1}{6}}} \end{pmatrix} \quad (\text{B.126})$$

with the function Y being the solution of an ordinary differential equation given in (B.65). The explicit form of the solution for Y can be written as

$$Y = \frac{Y_0 e^{\frac{\sigma}{6}}}{(g_1^2 e^{8\sigma} - 9m^2)^{\frac{1}{12}}} \quad (\text{B.127})$$

in which Y_0 is an integration constant.

It should be noted that the dilaton $\sigma(r)$ is not determined by the BPS equations as in [52, 53, 54, 55, 57] due to general covariance. We will use this freedom to fix the form of the solution for σ to be

$$\sigma = \frac{1}{8} \ln r. \quad (\text{B.128})$$

C Derivation of $SO(2)_{\text{diag}}$ symmetric solution

Similar to the previous section, in this appendix, we provide the derivation for the $AdS_4 \times \Sigma$ solution with $SO(2)_{\text{diag}}$ symmetry. We still use the same ansatz for the six-dimensional metric (B.1) as well as the two $SO(2)$ gauge fields (B.5). However, to implement the $SO(2)_{\text{diag}}$ symmetry generated by $J^{12} + \tilde{J}^{12}$, the two $SO(2)$ gauge fields are related by

$$g_1 A_1 = -g_2 A_2. \quad (\text{C.1})$$

We can also consistently set $B_{\mu\nu} = 0$ as in the previous case. For scalar fields, we have the dilaton $\sigma = \sigma(r)$ and the four $SO(2)_{\text{diag}}$ singlet scalars from the $SO(4,3)/(SO(4) \times SO(3))$ coset with the coset representative given in (4.1).

With all these, the field equation for the two-form field (A.1) becomes

$$0 = \frac{m e^{-2\sigma}}{2g_2} A'_1 (g_1 \sinh 2\phi_0 \cosh^2 \phi_2 + g_2 \sinh \phi_0 \sinh 2\phi_2) \quad (\text{C.2})$$

which implies ϕ_0 has to vanish in this case. For the vector field equation (A.2), the radial component ($\mu = r$) gives rise to the following non-trivial relation between the two scalar fields

$$\phi'_3 = \frac{\phi'_1}{2} \coth 2\phi_1 \sinh 4\phi_3 \quad (\text{C.3})$$

while the azimuthal component ($\mu = \theta$), with $A_2 = -\frac{g_1}{g_2} A_1$ and $\phi_0 = 0$, gives rise to

$$A''_1 = - \left(\frac{4f'}{f} - \frac{h'_1}{h_1} - \frac{h'_2}{h_2} - 4\sigma' + \frac{4g_1 \phi'_2}{g_2} \right) \frac{A'_1}{2}, \quad (\text{C.4})$$

$$A''_1 = - \left(\frac{4f'}{f} - \frac{h'_1}{h_1} - \frac{h'_2}{h_2} - 4\sigma' + \frac{4g_2 \phi'_2}{g_1} \right) \frac{A'_1}{2}. \quad (\text{C.5})$$

Consistency between the two equations for $\phi'_2 \neq 0$ requires

$$g_2 = \pm g_1 \quad (\text{C.6})$$

leading to a differential equation for the $A_1(r)$ function

$$A''_1 = - \left(\frac{4f'}{f} - \frac{h'_1}{h_1} - \frac{h'_2}{h_2} - 4\sigma' \pm 4\phi'_2 \right) \frac{A'_1}{2}. \quad (\text{C.7})$$

The plus/minus sign corresponds to that in the relation (C.6). From now on, we will choose $g_2 = -g_1$ for definiteness. This choice leads to $A_1 = A_2$, and equation (C.7) can be solved by

$$A'_1 = A'_2 = be^{2(\sigma+\phi_2)} \frac{\sqrt{h_1 h_2}}{f^2} \quad (\text{C.8})$$

where b is an integration constant.

Repeating the same analysis as in the previous case, we find that the BPS equations take the same form as in (B.33) to (B.41) when written in terms of the fermion-shift matrices. In particular, consistency still requires the following components $T_{[AB]}$, $S_{(AB)}$, and $N_{(AB)}$ to vanish in order to obtain non-trivial solutions. With the coset representative given in (4.1), non-vanishing components of these matrices take the following form

$$S_{(AB)} = -\frac{\sinh \phi_0}{4e^{3\sigma}} \left[m \sinh \phi_2 + \frac{g_1}{2} e^{4\sigma} (\cosh 2\phi_1 \cosh 2\phi_3 - 1) \right] \sigma_{AB}^3, \quad (\text{C.9})$$

$$N_{(AB)} = \frac{\sinh \phi_0}{8e^{3\sigma}} \left[6m \sinh \phi_2 - g_1 e^{4\sigma} (\cosh 2\phi_1 \cosh 2\phi_3 - 1) \right] \sigma_{AB}^3, \quad (\text{C.10})$$

$$T_{[AB]} = \frac{A'_1}{2} \sinh \phi_0 \epsilon_{AB}. \quad (\text{C.11})$$

We again see that ϕ_0 must vanish as previously required by the two-form field equation. Moreover, the quantities P_r^I , D^I_i , and $(L^{-1})^I_0$ also vanish when $\phi_0 = 0$. As a result, the BPS equation (B.38) is identically satisfied, and all the relevant BPS equations take the same form as those given in (B.53) to (B.57) with the only difference being in the explicit form of the remaining components of the fermion-shift matrices. With $\phi_0 = 0$, $g_2 = -g_1$, and $A'_1 = A'_2$, these components are given by

$$S_{[AB]} = \frac{1}{8} e^{3\sigma} \left[g_1 e^{4\sigma} (e^{-\phi_2} \cosh 2\phi_1 \cosh 2\phi_3 + e^{\phi_2}) + 2m \right] \epsilon_{AB}, \quad (\text{C.12})$$

$$N_{[AB]} = \frac{1}{8} e^{3\sigma} \left[g_1 e^{4\sigma} (e^{-\phi_2} \cosh 2\phi_1 \cosh 2\phi_3 + e^{\phi_2}) - 6m \right] \epsilon_{AB}, \quad (\text{C.13})$$

$$T_{(AB)} = \frac{A'_1 e^{-\phi_2}}{2\sqrt{h_1 h_2}} \sigma_{AB}^3 \quad (\text{C.14})$$

together with

$$P_r^I = \phi'_1 \cosh 2\phi_3 (\delta_1^I \delta_{i1} + \delta_2^I \delta_{i2}) - \phi'_3 (\delta_1^I \delta_{i2} - \delta_2^I \delta_{i1}) + \phi'_2 \delta_3^I \delta_{i3}, \quad (\text{C.15})$$

$$C^I_i = g_1 e^{-\phi_2} \left[\sinh 2\phi_1 (\delta_1^I \delta_{i1} + \delta_2^I \delta_{i2}) + \cosh 2\phi_1 \sinh 2\phi_3 (\delta_1^I \delta_{i2} - \delta_2^I \delta_{i1}) \right. \\ \left. + (e^{2\phi_2} - \cosh 2\phi_1 \cosh 2\phi_3) \right] \delta_3^I \delta_{i3}, \quad (\text{C.16})$$

$$T^I = \frac{A'_1 e^{-\phi_2}}{2\sqrt{h_1 h_2}} \delta_3^I. \quad (\text{C.17})$$

Substituting these quantities in (B.53) to (B.57), we finally obtained, after imposing the projector (B.50),

$$0 = \frac{s}{\sqrt{f}}\eta + \frac{1}{\sqrt{h_1}} \left(\frac{f'}{2f} + \sigma' \right) (i\sigma_2)\eta + 2me^{-3\sigma}\sigma_3\eta, \quad (\text{C.18})$$

$$0 = \frac{1}{\sqrt{h_1}}\partial_r\eta + \frac{1}{8}e^{3\sigma} [g_1e^{4\sigma} (e^{-\phi_2} \cosh 2\phi_1 \cosh 2\phi_3 + e^{\phi_2}) + 2m] \sigma_1\eta + \frac{3}{8}e^{-\sigma}\mathbb{F}_1(i\sigma_2)\eta, \quad (\text{C.19})$$

$$0 = \frac{g_1\hat{A}_1}{\sqrt{h_2}}\sigma_1\eta + \frac{1}{\sqrt{h_1}} \left(\frac{h'_2}{2h_2} + \sigma' \right) (i\sigma_2)\eta + 2me^{-3\sigma}\sigma_3\eta - e^{-\sigma}\mathbb{F}_1\eta, \quad (\text{C.20})$$

$$0 = \frac{\sigma'}{\sqrt{h_1}}\sigma_3\eta - \frac{1}{4}e^{3\sigma} [g_1e^{4\sigma} (e^{-\phi_2} \cosh 2\phi_1 \cosh 2\phi_3 + e^{\phi_2}) - 6m] (i\sigma_2)\eta + \frac{1}{4}e^{-\sigma}\mathbb{F}_1\sigma_1\eta, \quad (\text{C.21})$$

$$0 = \frac{\phi'_2}{\sqrt{h_1}}\sigma_3\eta - g_1e^{\sigma-\phi_2} (e^{2\phi_2} - \cosh 2\phi_1 \cosh 2\phi_3) (i\sigma_2)\eta + e^{-\sigma}\mathbb{F}_2\sigma_1\eta, \quad (\text{C.22})$$

$$0 = \frac{\phi'_1}{\sqrt{h_1}}\sigma_3\eta - g_1e^{\sigma-\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3 (i\sigma_2)\eta, \quad (\text{C.23})$$

$$0 = \frac{\phi'_3}{\sqrt{h_1}}\sigma_3\eta - g_1e^{\sigma-\phi_2} \cosh 2\phi_1 \sinh 2\phi_3 (i\sigma_2)\eta \quad (\text{C.24})$$

with

$$\mathbb{F}_1 = \mathbb{F}_2 = \frac{A'_1 e^{-\phi_2}}{2\sqrt{h_1 h_2}}. \quad (\text{C.25})$$

Note that (C.23) and (C.24) can be mapped to each other through the relation (C.3). Therefore, in this case, there are only five independent equations of the form $M^{(x)}\eta = 0$ where $x = 1, 2, 3, 4, 5$ respectively labels the BPS equations in (C.18), (C.20), (C.21), (C.22), and (C.23). On the other hand, equation (C.19) determines the explicit form of $\eta(r, \theta)$ as in the previous case.

After defining $\mathcal{A}^{xy}$, $\mathcal{B}^{xy}$, and $\mathcal{C}^{xy}$ matrices as in the previous case, we find that the vanishing of $\mathcal{A}^{55}$ gives the following condition

$$0 = \frac{(\phi'_1)^2}{h_1} - g_1^2 e^{2(\sigma-\phi_2)} \sinh^2 2\phi_1 \operatorname{sech}^2 2\phi_3. \quad (\text{C.26})$$

The symmetric part $\mathcal{A}^{\tilde{x}5} + \mathcal{A}^{5\tilde{x}} = 0$ with $\tilde{x} = 1, 2, 3, 4$ gives

$$0 = \frac{e^{-\sigma-\phi_2}}{\sqrt{h_1}} \left[g_1 e^{4\sigma} \sinh 2\phi_1 \operatorname{sech} 2\phi_3 \left(\frac{f'}{f} + 2\sigma' \right) + 4me^{\phi_2} \phi'_1 \right], \quad (\text{C.27})$$

$$0 = \frac{e^{-\sigma-\phi_2}}{\sqrt{h_1}} \left[g_1 e^{4\sigma} \sinh 2\phi_1 \operatorname{sech} 2\phi_3 \left(\frac{h'_2}{h_2} + 2\sigma' \right) + 4me^{\phi_2} \phi'_1 \right], \quad (\text{C.28})$$

$$0 = \frac{\sigma' \phi'_1}{h_1} - \frac{g_1 [2e^{\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3 (g_1 e^{4\sigma+\phi_2} - 6m) + g_1 e^{4\sigma} \sinh 4\phi_1]}{8e^{2(\sigma+\phi_2)}}, \quad (\text{C.29})$$

$$0 = \frac{2\phi'_1 \phi'_2}{h_1} - g_1^2 e^{2(\sigma-\phi_2)} (2e^{2\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3 - \sinh 4\phi_1). \quad (\text{C.30})$$

The antisymmetric part $\mathcal{A}^{\tilde{x}5} - \mathcal{A}^{5\tilde{x}} = 0$ yields

$$0 = \frac{s\phi'_1}{\sqrt{f}h_1}, \quad (\text{C.31})$$

$$0 = \frac{\phi'_1}{\sqrt{h_1}} e^{-\sigma} \mathbb{F}_1 - \frac{g_1^2 \hat{A}_1}{\sqrt{h_2}} e^{\sigma-\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3, \quad (\text{C.32})$$

$$0 = g_1 \mathbb{F}_1 e^{-\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3, \quad (\text{C.33})$$

$$0 = g_1 \mathbb{F}_2 e^{-\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3. \quad (\text{C.34})$$

From $\mathcal{B}^{\tilde{x}5} + \mathcal{C}^{\tilde{x}5} = 0$, we find

$$0 = \frac{sg_1}{\sqrt{f}} e^{\sigma-\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3, \quad (\text{C.35})$$

$$0 = \frac{g_1 \hat{A}_1 \phi'_1}{\sqrt{h_1 h_2}} - g_1 \mathbb{F}_1 e^{-\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3, \quad (\text{C.36})$$

$$0 = \frac{\mathbb{F}_1 e^{-\sigma} \phi'_1}{\sqrt{h_1}}, \quad (\text{C.37})$$

$$0 = \frac{\mathbb{F}_2 e^{-\sigma} \phi'_1}{\sqrt{h_1}}. \quad (\text{C.38})$$

Finally, $\mathcal{B}^{\tilde{x}5} - \mathcal{C}^{\tilde{x}5} = 0$ gives

$$0 = \frac{\phi'_1}{h_1} \left(\frac{f'}{f} + 2\sigma' \right) + 4g_1 m e^{-2\sigma-\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3, \quad (\text{C.39})$$

$$0 = \frac{\phi'_1}{h_1} \left(\frac{h'_2}{h_2} + 2\sigma' \right) + 4g_1 m e^{-2\sigma-\phi_2} \sinh 2\phi_1 \operatorname{sech} 2\phi_3, \quad (\text{C.40})$$

$$0 = \frac{\phi'_1}{4\sigma'} \left[(\cosh 2\phi_1 \cosh 2\phi_3 + e^{2\phi_2}) - \frac{6m e^{\phi_2}}{g_1 e^{4\sigma}} \right] - \sinh 2\phi_1 \operatorname{sech} 2\phi_3, \quad (\text{C.41})$$

$$0 = \frac{2g_1 e^{\sigma-\phi_2}}{\sqrt{h_1}} [\phi'_1 (e^{2\phi_2} - \cosh 2\phi_1 \cosh 2\phi_3) - \sinh 2\phi_1 \phi'_2 \operatorname{sech} 2\phi_3]. \quad (\text{C.42})$$

From these results, we can readily see that setting $\phi_1 = 0$ satisfies all of these conditions without any additional constraint. More over, from (C.3), we also find that $\phi'_3 = 0$ for $\phi'_1 = 0$ giving rise to constant ϕ_3 that can be set to zero. With $\phi_1 = \phi_3 = 0$, we find that $S_{[AB]}$, $N_{[AB]}$, $P_r^I{}_i$, and $C^I{}_i$ from (C.12), (C.13), (C.15), and (C.16) reduce to (B.58), (B.59), (B.61), and (B.62), respectively. Accordingly, the non-trivial BPS equations as well as the conditions resulted from the vanishing of $\mathcal{A}^{\tilde{x}\tilde{y}}$, $\mathcal{B}^{\tilde{x}\tilde{y}}$, and $\mathcal{C}^{\tilde{x}\tilde{y}}$ are also the same as in (B.64) to

(B.68) and (B.74) to (B.101) obtained in the previous case with $\mathbb{F}_1$ and $\mathbb{F}_2$ given in (C.25).

As a result, the $SO(2)_{\text{diag}}$ symmetric solution is then given by

$$f = \frac{e^{\frac{10\sigma}{3}} (bg_1 m)^{\frac{2}{3}}}{(s(g_1^2 e^{8\sigma} - 9m^2))^{\frac{2}{3}}}, \quad (\text{C.43})$$

$$h_1 = \frac{576m^4 e^{6\sigma} (bg_1 m)^{\frac{2}{3}} (\sigma')^2}{(g_1^2 e^{8\sigma} - 9m^2)^2 \left[4m^2 (bg_1 m)^{\frac{2}{3}} - e^{\frac{8\sigma}{3}} (s(g_1^2 e^{8\sigma} - 9m^2))^{\frac{2}{3}} \right]}, \quad (\text{C.44})$$

$$h_2 = \frac{9C^2 g_1^2 m^2 e^{\frac{10\sigma}{3}}}{(s(g_1^2 e^{8\sigma} - 9m^2))^{\frac{2}{3}}} \left[4m^2 (bg_1 m)^{\frac{2}{3}} - e^{\frac{8\sigma}{3}} (s(g_1^2 e^{8\sigma} - 9m^2))^{\frac{2}{3}} \right], \quad (\text{C.45})$$

$$\phi_2 = 4\sigma + \ln \left[\frac{g_1}{3m} \right], \quad (\text{C.46})$$

$$A_1 = C(g_1^2 e^{8\sigma} - 3m^2) - \frac{2q}{g_1} \quad (\text{C.47})$$

where C is an integration constant. It can also be verified that all of the BPS conditions and the field equations are satisfied for

$$bg_1 m > 0, \quad s(g_1^2 e^{8\sigma} - 9m^2) > 0, \quad \frac{g_1}{3m} > 0. \quad (\text{C.48})$$

Finally, solving equation (C.19), we find the Killing spinor given by

$$\eta = \frac{e^{i\theta q} Y_0 e^{\frac{5\sigma}{6}}}{(g_1^2 e^{8\sigma} - 9m^2)^{\frac{1}{6}}} \begin{pmatrix} \sqrt{2ms(bg_1 m)^{\frac{1}{3}} - e^{\frac{4\sigma}{3}} (s(g_1^2 e^{8\sigma} - 9m^2))^{\frac{1}{3}}} \\ \sqrt{2ms(bg_1 m)^{\frac{1}{3}} + e^{\frac{4\sigma}{3}} (s(g_1^2 e^{8\sigma} - 9m^2))^{\frac{1}{3}}} \end{pmatrix}. \quad (\text{C.49})$$

As in the previous case, all functions are determined in terms of the dilaton $\sigma(r)$. In section 4, we have chosen the solution for σ as in (B.128).

D Relations to spindle solutions

$AdS_4 \times \Sigma$ solutions with Σ being a spindle and a topological disk take the same local form but with different global extensions. A general classification of $AdS_4 \times \Sigma$ solutions within the $U(1) \times U(1)$ subsector of matter-coupled $F(4)$ gauged supergravity has been given in [43]. In this appendix, we show that the solutions given in this paper are related to the solutions describing D4-branes wrapped on a spindle studied in [50]. In addition, we will show that the solutions presented here extend the known spindle and disk solutions classified in [43].

We first recall that the $AdS_4 \times \Sigma$ solution for Σ being a spindle given in [50] consists of the metric, gauge fields, and scalars as in this paper. In the

notation of [43], the solution can be written as

$$\begin{aligned} ds_6^2 &= (y^2 h_1 h_2)^{1/4} \left(ds_{AdS_4}^2 + \frac{y^2}{F} dy^2 + \frac{F}{h_1 h_2} dz^2 \right), \\ \mathcal{A}_i &= \left(\alpha_i - \frac{y^3}{h_i} \right) dz, \quad X_i = \frac{(y^2 h_1 h_2)^{3/8}}{h_i} \end{aligned} \quad (D.1)$$

in which

$$F(y) = m^2 h_1 h_2 - y^4 \quad \text{and} \quad h_i(y) = \frac{2g}{3m} y^3 + q_i \quad (D.2)$$

with α_i and q_i for $i = 1, 2$ being constants. Following appendix A.1 of [50], these quantities can be expressed in the present notation as

$$\begin{aligned} X_1 &= e^{\sigma+\phi_2}, \quad X_2 = e^{\sigma-\phi_2}, \quad g = g_1, \quad m \rightarrow \frac{m}{2}, \\ \mathcal{A}_1 &= \frac{1}{2}(A^3 + A^6), \quad \mathcal{A}_2 = \frac{1}{2}(A^3 - A^6). \end{aligned} \quad (D.3)$$

D.1 $SO(2) \times SO(2)$ symmetric solution

From the $SO(2) \times SO(2)$ symmetric solution given in section 3, we find

$$X_1 = r^{\frac{1}{8}} \left(G\sqrt{r} + \sqrt{G^2 r - 1} \right), \quad (D.4)$$

$$X_2 = r^{\frac{1}{8}} \left(G\sqrt{r} - \sqrt{G^2 r - 1} \right), \quad (D.5)$$

$$\mathcal{A}_1 = \left[\frac{2\mathcal{C}}{m} \left(Gr + \sqrt{r(G^2 r - 1)} + \frac{c}{2} - 2 \left(\frac{q + 2\mathcal{C}}{3Gm} \right) \right) \right] dz, \quad (D.6)$$

$$\mathcal{A}_2 = \left[\frac{2\mathcal{C}}{m} \left(Gr - \sqrt{r(G^2 r - 1)} \right) - \frac{c}{2} - 2 \left(\frac{q + 2\mathcal{C}}{3Gm} \right) \right] dz. \quad (D.7)$$

Comparing the metric in (D.1) with (3.8) and (3.9) yields the coordinate transformations

$$r = \frac{m^6 y^6}{G^2 m^6 y^6 - B^3} \quad \text{and} \quad \theta = \frac{mz}{2\mathcal{C}} \quad (D.8)$$

that map the spindle metric in (D.1) to the disk solution in (3.8) as follows

$$F(y) = \frac{B^2 \left[B - r^{\frac{2}{3}} (G^2 r - 1)^{\frac{1}{3}} \right]}{m^4 (G^2 r - 1)} = m^2 h_1 h_2 - y^4, \quad (D.9)$$

$$h_1(y) = \frac{B^{\frac{3}{2}}}{m^3 \left[G\sqrt{r(G^2 r - 1)} + G^2 r - 1 \right]} = Gy^3 - \frac{B^{\frac{3}{2}}}{m^3}, \quad (D.10)$$

$$h_2(y) = \frac{B^{\frac{3}{2}}}{m^3 \left[G\sqrt{r(G^2 r - 1)} - G^2 r + 1 \right]} = Gy^3 + \frac{B^{\frac{3}{2}}}{m^3}. \quad (D.11)$$

Recalling $W(r) = B - r^{\frac{2}{3}}(G^2 r - 1)^{\frac{1}{3}}$, we see that r_1 given in (3.16) is also a unique real root of $F(y)$ while, at the other endpoint $r = \frac{1}{G^2}$, the function $F(y)$ diverges. Moreover, comparing the functions h_1 and h_2 in (D.10) and (D.11) with (D.2), we find that the two constants q_i take the form

$$q_1 = -\frac{B^{\frac{3}{2}}}{m^3} \quad \text{and} \quad q_2 = \frac{B^{\frac{3}{2}}}{m^3}. \quad (\text{D.12})$$

For the sake of comparison to the spindle and previous disk solutions, it is convenient to define the following two parameters as in [43]

$$s = m^3(q_1 + q_2) \quad \text{and} \quad p = m^6 q_1 q_2. \quad (\text{D.13})$$

In terms of these parameters, we find that $SO(2) \times SO(2)$ symmetric solution given here has $s = 0$ and $p = -B^3$. Since B^3 is an arbitrary positive real parameter, see (3.11), this implies that the $SO(2) \times SO(2)$ symmetric solution found in this work is equivalent to the spindle solution in [50] with $s = 0$ and $p < 0$. For clarity, we have added a red line representing solutions with $s = 0$ and $p < 0$ to the plot of the domains of validity on the (s, p) plane given in [43] as shown in figure 9. As pointed out in [43], the spindle solution of [50] exists in the region R_1 while the disk solution of [54] exists in the regions R_2 and R_3 . On the other hand, the region R_4 contains non-compact domain-wall solutions with infinite holographic free energy. Consequently, we conclude that the $SO(2) \times SO(2)$ symmetric solution found here represents a novel case that interpolates between a single root of $F(y) = 0$ and a singularity.

D.2 $SO(2)_{\text{diag}}$ symmetric solution

For the $SO(2)_{\text{diag}}$ symmetric solution studied in section 4, we have

$$\begin{aligned} X_1 &= Gr^{\frac{5}{8}}, & X_2 &= \frac{1}{Gr^{\frac{3}{8}}}, \\ \mathcal{A}_1 &= \left(-\frac{2(2q + \mathcal{C})}{3Gm} + \frac{2G\mathcal{C}r}{m} \right) dz, & \mathcal{A}_2 &= 0. \end{aligned} \quad (\text{D.14})$$

In this case, the metric of the spindle solution given in (D.1) and that of the disk given in (4.3) and (4.4) are equivalent after changing the radial and azimuthal coordinates as follows

$$r = \frac{km^3 y^3}{kG^2 m^3 y^3 - B^3} \quad \text{and} \quad \theta = \frac{mz}{2\mathcal{C}}. \quad (\text{D.15})$$

To avoid confusion with the parameter s defined in (D.13), we have temporarily changed the sign parameter s to k in the solution.

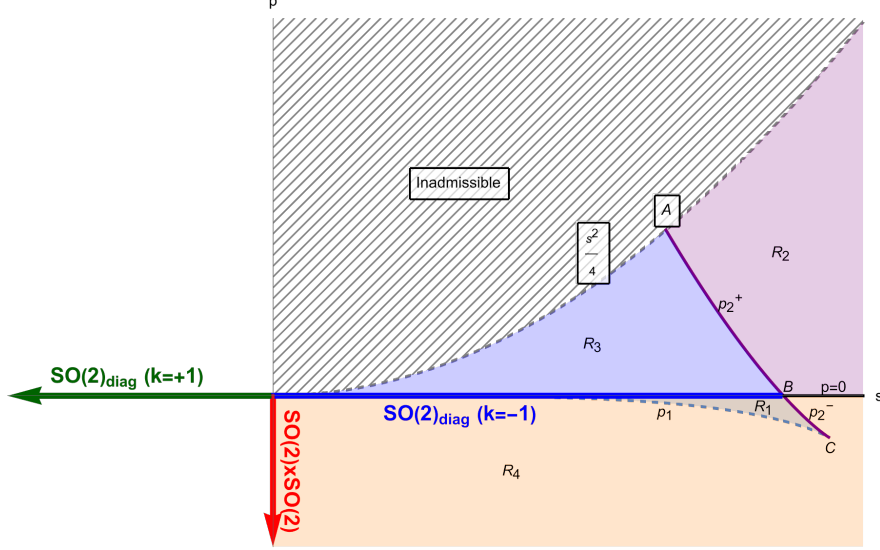


Figure 9: Various domains of validity for $AdS_4 \times \Sigma$ solutions in (s, p) space. This plot is the same as that given in [43] with four regions R_1 , R_2 , R_3 , and R_4 corresponding to different global completions. The regions are separated by different lines with the two non-trivial lines $p = p_1(s)$ and $p = p_2(s)$ given explicitly in [43]. A superscript $\pm$ on p_2 denotes whether this line lies above or below the s axis. There are three distinguished points arising from the intersection of the boundary lines shown by $A = \left(\frac{8}{27}, \frac{16}{729}\right)$, $B = \left(\frac{2}{3\sqrt{3}}, 0\right)$, and $C = \left(\frac{8\sqrt{2}}{27}, -\frac{4}{729}\right)$.

The various functions appearing in the metric are now given by

$$F(y) = \frac{B^2 r \left[B - r^{\frac{1}{3}} (k(G^2 r - 1))^{\frac{2}{3}} \right]}{m^4 (G^2 r - 1)} = m^2 h_1 h_2 - y^4, \quad (D.16)$$

$$h_1(y) = \frac{B^{\frac{2}{3}}}{k G m^3 (G^2 r - 1)} = G y^3 - \frac{B^{\frac{2}{3}}}{k G m^3}, \quad (D.17)$$

$$h_2(y) = \frac{B^{\frac{2}{3}} G r}{k m^3 (G^2 r - 1)} = G y^3. \quad (D.18)$$

As $W(r) = B - r^{\frac{1}{3}} (k(G^2 r - 1))^{\frac{2}{3}}$, all three roots determined from $W(r) = 0$ in (4.12) and (4.13) simultaneously solve $F(y) = 0$. Furthermore, it is straightforward to verify that $r = 0$ constitutes another root of $F(y)$. As in the $SO(2) \times SO(2)$ case, $r = \frac{1}{G^2}$ leads to a divergence in $F(y)$. In this case, we find $q_2 = 0$ while the parameter q_1 is given by

$$q_1 = -\frac{B^{\frac{3}{2}}}{k G m^3}. \quad (D.19)$$

From (D.13), we find that the $SO(2)_{\text{diag}}$ symmetric disk solution given here is equivalent to the spindle solution in [50] with $s = -\frac{B^{\frac{3}{2}}}{k G}$ and $p = 0$.

However, in section 4, we find two different behaviors in the solution depending on whether $k = +1$ or $k = -1$. For $k = +1$, a regular solution exists when $B > 0$, $G > 0$, and $m > 0$. This corresponds to the green line ($s < 0$, $p = 0$) shown in figure 9 which represents a novel case discovered in this work. On the other hand, for $k = -1$, we find a solution with an $\mathbb{R}^2/\mathbb{Z}_l$ orbifold singularity only when $0 < B < \frac{2^{\frac{2}{3}}}{3G^{\frac{2}{3}}}$, $G > 0$, and $m > 0$. Setting $2g = 3m$ in [43] which leads to $G = 1$ here, we find that this case corresponds to the blue line ($0 < s < \frac{2}{3\sqrt{3}}$, $p = 0$) in figure 9.

As pointed out in [43], there are two types of disk solutions with and without enhanced symmetry. An example of the latter is the solution found in [54] and the $SO(2) \times SO(2)$ symmetric solution discussed above. The solution on the blue line in figure 9 closely resembles the disk with enhanced symmetry. However, only solutions interpolating between zero and a single non-vanishing root have been examined in [43]. This type of solutions should be identified with $SO(2)_{\text{diag}}$ solution found here in the range $0 < r < r_+$ corresponding to the left region of the numerical plots shown in figure 5. Apart from this, the present work suggests the existence of another type of solutions interpolating between another root and a singularity corresponding to a distinct range $r_- < r < \frac{1}{G^2}$.

E Uplift to massive type IIA theory

In this appendix, we give a truncation ansatz of massive type IIA theory on a half four-sphere as given in [43], see also [50] and [72] for earlier results. This truncation leads to $U(1) \times U(1)$ invariant sector of $F(4)$ gauged supergravity coupled to vector multiplets. All solutions from $SO(3) \times SO(3)$ gauged supergravity studied in this paper lie within this invariant sector and can be uplifted to ten dimensions by using the aforementioned truncation ansatz, see [50] for the uplifted solutions describing D4-branes wrapped on a Riemann surface and a spindle.

In the present convention, the metric ansatz in the string frame is given by

$$ds^2 = \lambda^2 \mu_0^{-\frac{1}{3}} (X_1 X_2)^{-\frac{1}{4}} \Delta^{\frac{1}{2}} \left\{ ds_6^2 + g_1^{-2} \Delta^{-1} [X_0^{-1} d\mu_0^2 + X_1^{-1} (d\mu_1^2 + \mu_1^2 D\varphi_1^2) + X_2^{-1} (d\mu_2^2 + \mu_2^2 D\varphi_2^2)] \right\}. \quad (\text{E.1})$$

X_1 and X_2 are related to the dilaton σ and $SO(2) \times SO(2)$ singlet scalar ϕ_2 as in (D.3). On the other hand, X_0 is defined by

$$X_0 = (X_1 X_2)^{-\frac{3}{2}} = e^{-3\sigma}. \quad (\text{E.2})$$

The constrained coordinates μ_a , $a = 0, 1, 2$, satisfy

$$\mu_0^2 + \mu_1^2 + \mu_2^2 = 1 \quad (\text{E.3})$$

and can be chosen to be

$$\mu_0 = \cos \xi, \quad \mu_1 = \sin \xi \sin \eta, \quad \mu_2 = \sin \xi \cos \eta \quad (\text{E.4})$$

with $\eta \in [0, \frac{\pi}{2}]$ and $\xi \in [0, \frac{\pi}{2}]$. The two angular coordinates φ_1 and φ_2 have period 2π with $D\varphi_i = d\varphi_i - g_1 \mathcal{A}_i$, $i = 1, 2$. The two gauge fields $\mathcal{A}_{1,2}$ are again related to the two gauge fields A^3 and A^6 as in (D.3).

The ten-dimensional dilaton is given by

$$e^\Phi = \lambda^2 \mu_0^{-\frac{5}{6}} \Delta^{\frac{1}{4}} (X_1 X_2)^{-\frac{5}{8}}. \quad (\text{E.5})$$

The zero-form Romans mass is

$$F_{(0)} = \frac{m}{\lambda^3} = \frac{g_1}{3\lambda^3} \quad (\text{E.6})$$

Since the two-form field vanishes in all of the solutions considered here, we will set the two-form field to zero in all subsequent expressions. This results in the following four-form field strength

$$\begin{aligned} F_{(4)} = & \sqrt{\frac{g_1}{3m}} \frac{\lambda \mu_0^{-\frac{2}{3}}}{2g_1^3 \Delta} \left\{ 2U \mu_1 \mu_2 \Delta^{-1} d\mu_1 \wedge d\mu_2 \wedge D\varphi_1 \wedge D\varphi_2 \right. \\ & - 2X_1 X_2 \Delta^{-1} \left[\frac{\mu_1 \mu_2 \mu_0^2 X_0^2}{X_1 X_2} \left(\mu_1 d\frac{X_1}{X_2} \wedge d\mu_2 - \mu_2 d\frac{X_2}{X_1} \wedge d\mu_1 \right) \right. \\ & \left. \left. - \mu_1^2 \mu_2^2 \mu_0 d \ln \frac{X_1}{X_2} \wedge d\mu_0 \right] \wedge D\varphi_1 \wedge D\varphi_2 \right\} \\ & + \sqrt{\frac{g_1}{3m}} \frac{\lambda \mu_0^{-\frac{2}{3}}}{g_1^2 \Delta} \left\{ \mathcal{F}_1 \wedge [\mu_1 \mu_2^2 X_2 d\mu_1 + \mu_2 (X_0 \mu_0^2 + X_2 \mu_2^2) d\mu_2] \wedge D\varphi_2 \right. \\ & \left. + \mathcal{F}_2 \wedge [\mu_2 \mu_1^2 X_1 d\mu_2 + \mu_1 (X_0 \mu_0^2 + X_1 \mu_1^2) d\mu_1] \wedge D\varphi_1 \right\}. \quad (\text{E.7}) \end{aligned}$$

All the remaining fields in massive type IIA theory vanish. In all the above expressions, we have included a parameter λ such that the uplifted ten-dimensional solutions can be defined globally with all the fluxes properly quantized along the internal compact dimensions, see more detail in [50].

The $AdS_4 \times \Sigma$ geometries are expected to holographically described three-dimensional SCFTs arising from compactifications of $N = 2$ SCFTs in five dimensions. The holographic free energy of the three-dimensional SCFTs can be computed by writing the ten-dimensional metric in string frame as

$$ds_{10}^2 = e^{2\mathbf{A}} (ds_{AdS_4}^2 + ds_{M_6}^2). \quad (\text{E.8})$$

As in [71] and [50], the free energy is then given by an integral over the internal space M_6 of the form

$$\mathbf{F} = \frac{16\pi^3 \lambda^4}{(2\pi \ell_s)^8} \int_{M_6} e^{8\mathbf{A} - 2\Phi} \text{vol}_{M_6}. \quad (\text{E.9})$$

For the six-dimensional metric given in (B.1), we find the uplifted ten-dimensional metric of the form

$$ds_{10}^2 = \cos^{-\frac{1}{3}} \xi e^{-\frac{\sigma}{2}} \Delta^{\frac{1}{2}} f \left[ds_{AdS_4}^2 + \frac{h_1}{f} dr^2 + \frac{e^{-\sigma} \Delta^{-1}}{g_1^2 f} ds_{M_2}^2 + ds_{M_3}^2 \right] \quad (E.10)$$

in which the metrics on M_2 and M_3 submanifolds are given respectively by

$$ds_{M_2}^2 = \sin^2 \xi \left(e^{-\phi_2} \cos^2 \eta + e^{\phi_2} \sin^2 \eta \right) d\eta^2 - \sin 2\xi \sin 2\eta \sinh \phi_2 d\xi d\eta + \left[e^{4\sigma} \sin^2 \xi + \cos^2 \xi (e^{-\phi_2} \sin^2 \eta + e^{\phi_2} \cos^2 \eta) \right] d\xi^2 \quad (E.11)$$

and

$$\begin{aligned} ds_{M_3}^2 &= \frac{h_2}{f} d\theta^2 + \frac{e^{-\sigma} \Delta^{-1} \sin^2 \xi}{g_1^2 f} \left[e^{\phi_2} \cos^2 \eta (d\varphi_2 - g_1 \mathcal{A}_{2\theta} d\theta)^2 \right. \\ &\quad \left. + e^{-\phi_2} \sin^2 \eta (d\varphi_1 - g_1 \mathcal{A}_{1\theta} d\theta)^2 \right] \\ &= \frac{e^{-\sigma} \Delta^{-1}}{g_1^2 f} \left[g_1^2 h_2 e^{\sigma} \Delta + \sin^2 \xi \left(e^{-\phi_2} \mathcal{A}_{1\theta}^2 \sin^2 \eta + e^{\phi_2} \mathcal{A}_{2\theta}^2 \cos^2 \eta \right) \right] d\theta^2 \\ &\quad + \frac{e^{-\sigma} \Delta^{-1} \sin^2 \xi}{g_1^2 f} \left[e^{-\phi_2} \sin^2 \eta d\varphi_1^2 + e^{\phi_2} \cos^2 \eta d\varphi_2^2 \right. \\ &\quad \left. - 2g_1 \left(e^{-\phi_2} \mathcal{A}_{1\theta} \sin^2 \eta d\theta d\varphi_1 + e^{\phi_2} \mathcal{A}_{2\theta} \cos^2 \eta d\theta d\varphi_2 \right) \right]. \end{aligned} \quad (E.12)$$

We can then identify

$$e^{\mathbf{A}} = \cos^{-\frac{1}{6}} \xi e^{-\frac{\sigma}{4}} \Delta^{\frac{1}{4}} f^{\frac{1}{2}} \quad (E.13)$$

and find the volume form on M_6 as

$$\begin{aligned} \text{vol}_{M_6} &= \sqrt{\frac{h_1}{f}} \left(\frac{\Delta^{-1} e^{-\sigma}}{g_1^2 f} \right) \sqrt{\det g_{M_2} \det g_{M_3}} dr \wedge d\eta \wedge d\xi \wedge d\theta \wedge d\varphi_1 \wedge d\varphi_2 \\ &= \frac{\sqrt{h_1 h_2}}{g_1^3 f^3} \Delta^{-\frac{3}{2}} e^{-\frac{\sigma}{2}} \sin^3 \xi \cos \eta \sin \eta dr \wedge d\eta \wedge d\xi \wedge d\theta \wedge d\varphi_1 \wedge d\varphi_2 \end{aligned} \quad (E.14)$$

in which we have used the expression for the warp factor of the form

$$\Delta = e^{-3\sigma} \left[\cos^2 \xi + \sin^2 \xi e^{4\sigma - \phi_2} (\cos^2 \eta + e^{2\phi_2} \sin^2 \eta) \right]. \quad (E.15)$$

Using the explicit form of the ten-dimensional dilaton

$$e^{\Phi} = \cos^{-\frac{5}{6}} \xi \Delta^{\frac{1}{4}} e^{-\frac{5\sigma}{4}}, \quad (E.16)$$

we eventually obtain the free energy

$$\mathbf{F} = \frac{9\lambda^4}{80\pi^2 g_1^3 \ell_s^8} \int f \sqrt{h_1 h_2} dr. \quad (E.17)$$

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